

The Physical Clock as a Quotient

AASC Time Non-Primitivity II: Clock Use without a Primitive Clock

Internal Time and Clock Choice in Constrained Quantum Gravity

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Abstract

This is the second paper in the AASC time non-primitivity sequence. The AASC kernel is instantiated by necessity in every non-degenerate regime that supports admissibility-assessable, identity-preserving standing-bearing construction. It is not generated by clock success, clock selection, metric time, relational conditioning, or the quotient theorem proved here. Paper I, *The Arrow of Time as Standing Monotonicity* (DOI: [10.5281/zenodo.20678464](https://doi.org/10.5281/zenodo.20678464)), established the directionality consequence for regimes that remain kernel-faithful. The present paper treats the separate clock-carrier problem: when a variable, subsystem, field, modular flow, record sequence, semiclassical parameter, or relational conditioning construction has standing as a physical clock for a declared use.

AASC is a constraint formalism. Its primary operational mode here is exhaustion and impossibility. The contribution is a closure theorem over clock constructions rather than another clock construction: once the constrained domain, declared use, and kernel-faithful comparison class are fixed, every standing-relevant same-scope clock difference must enter evaluated classification or coherent retained continuation, and every attempted singular selection must terminate in the governance normal form.

Constrained quantum gravity admits many formal clock constructions: relational observables, deparametrized matter clocks, Page–Wootters conditioning, semiclassical parameters, modular flows, and record-ordered histories. The structural question is what constitutes the physical clock object and whether clock choice leaves any untyped same-scope residue once the domain and use are fixed.

A constrained clock-use package fixes the comparison class: boundary and admissibility typing, evaluated clock-use fragment, physical quotient object, readout and record burden, retained continuation, failure boundary, and anchor–tensor–skin role carrier. The AMetric closure theorem then exhausts every standing-relevant clock-use difference by evaluated classification or coherent retained-trajectory structure. The complete admitted clock-use probe family is defined over those loci, and canonical clock equivalence is agreement of every admitted probe in both definedness and value. The resulting physical clock is therefore not a preferred variable, subsystem label, gauge slice, or record token. It is the use-relative quotient class of certified same-scope clock-use behavior.

The capstone theorem combines clock-profile finality, quotient uniqueness, a five-branch physical normal form for clock-candidate pairs, and a separate ordered governance theorem for singular selection. Candidate pairs are theorem-object divergent, eligibility divergent, anchor divergent, quotient-profile divergent, or physically equivalent. A selection act is governed by declared non-trace role content, a previously declared singleton boundary trace, scope enrichment, representative privilege, unsupported class privilege, or an impossible authority-acquisition route.

The AMetric route-exhaustion theorem and the no-repair/no-generator/no-carrier apex closure eliminate any additional same-scope selector, parameter, second classifier, transfer, repair, or standing-lift route.

The canonical clock quotient for a declared (D, U) package may be empty, singleton, or physically plural inside one unique standing-governed admissible interior. Typed plurality is not residual ambiguity and does not authorize a preferred class. A scalar-field minisuperspace audit, boundary-time analysis, and classifications of material, semiclassical, Page–Wootters, modular, and histories clocks demonstrate realization, failure, plurality, and lawful fixation without restoring primitive external time. The theorem boundary leaves quantum-gravity dynamics, global clock existence, the Born rule, and preferred-clock construction in plural regimes to distinct or richer targets.

Keywords: AASC; constrained quantum gravity; problem of time; physical clock; clock quotient; internal time; relational time; clock choice; quotient finality; multiple-choice problem; selection governance; AMetric boundary; admissibility; standing.

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Roadmap and exact claim

Series dependency. The dependency order is fixed: kernel by necessity; downstream consequence by kernel-faithful specialization. The kernel is already in force wherever non-degenerate standing-bearing construction is present. Paper I derives the directionality consequence. Paper II derives the clock-use consequence. The present manuscript does not rederive standing monotonicity or generate admissibility, standing, reference, or irreversibility from clock success. A physical clock may order or condition content for a declared use, but it does not generate the kernel, the standing arrow, or any effective arrow.

Shared lock. Clock standing is not temporal direction; temporal direction is not clock selection.

Core thesis. For a fixed constrained domain D and declared clock use U , the physical clock is the canonical final quotient of certified same-scope clock-use behavior. Pairwise clock relations are exhausted by theorem-object divergence, eligibility divergence, anchor divergence, quotient-profile divergence, or equivalence. Singular selection is a separate governance act exhausted by declared role fixation, prior boundary-trace fixation, scope enrichment, representative privilege, unsupported class privilege, or impossible authority acquisition. No residual same-scope clock identity, relation, or selector authority survives outside the quotient and declared anchor structure.

The proof order is

fixed clock-use target $\implies$ complete admitted probe family,
 probe family $\implies$ canonical clock equivalence and quotient,
 constructional-behavior exhaustion $\implies$ profile sufficiency, profile necessity, and finality,
 final quotient $\implies$ clock-pair physical normal form,
 pair normal form plus impossibility closure $\implies$ clock-selection governance,
 clock-selection governance $\implies$ choice typing closure.

No clock theorem is used to establish the imported AASC kernel, admissibility interface, fixed-domain exhaustion, or AER quotient architecture. The kernel is instantiated by necessity in the non-degenerate target class; the manuscript begins after that closure and proves the clock-specific consequence for a kernel-faithful constrained domain.

A regime is **kernel-faithful** when its standing-bearing claims preserve admissibility, standing, reference, and irreversibility; preserve the distinction between admissible continuation and prohibited same-act repair; preserve target identity under lawful redescription; and introduce no clock, external parameter, probability, typicality rule, metric, representative, success criterion, or boundary trace as standing-generating authority before admission.

The central result is the Physical-Clock Quotient Finality and Choice-Closure Theorem in [theorem 15.5](#). Its prerequisites are the Clock-Profile Sufficiency and Necessity Theorem in [theorem 5.11](#), the Clock-Profile Finality Theorem in [theorem 5.12](#), and the Clock-Quotient Uniqueness Theorem in [theorem 5.13](#). The Clock-Pair Physical Normal-Form Theorem in [theorem 14.4](#) classifies candidate relations; the Clock-Selection Governance Theorem in [theorem 15.3](#) classifies attempted singular selections. The quotient-status corollary in [theorem 15.9](#) distinguishes no-clock, singleton-clock, and physically plural regimes. Saturated registry completion is stated separately in [theorem 15.12](#).

Established background. Canonical quantum gravity lacks ordinary external Schrödinger time; physical predictions may be relational; internal clocks can fail globally; different clock choices can produce inequivalent reductions or quantum theories; and semiclassical time is approximation-dependent. **Constraint-formalism result.** The manuscript identifies the physical clock with the final use-relative quotient of certified clock-use behavior; proves profile sufficiency, profile necessity, finality, and quotient uniqueness; separates candidate-pair relations from selection governance; and closes both by the AMetric comparison-class, route-exhaustion, and apex uniqueness results. The multiple-choice problem is thereby converted into the structure of the quotient: empty, singleton, or typed plurality inside one admissible interior.

Theorem boundary. Clock realization, a preferred class in plural regimes, a Born rule, a global quantum-gravity dynamics, and finite-registry completeness without saturation belong to distinct or richer targets. The clock-finality theorem neither requires nor implies them, and it does not backflow into the kernel or Paper I's standing-arrow result.

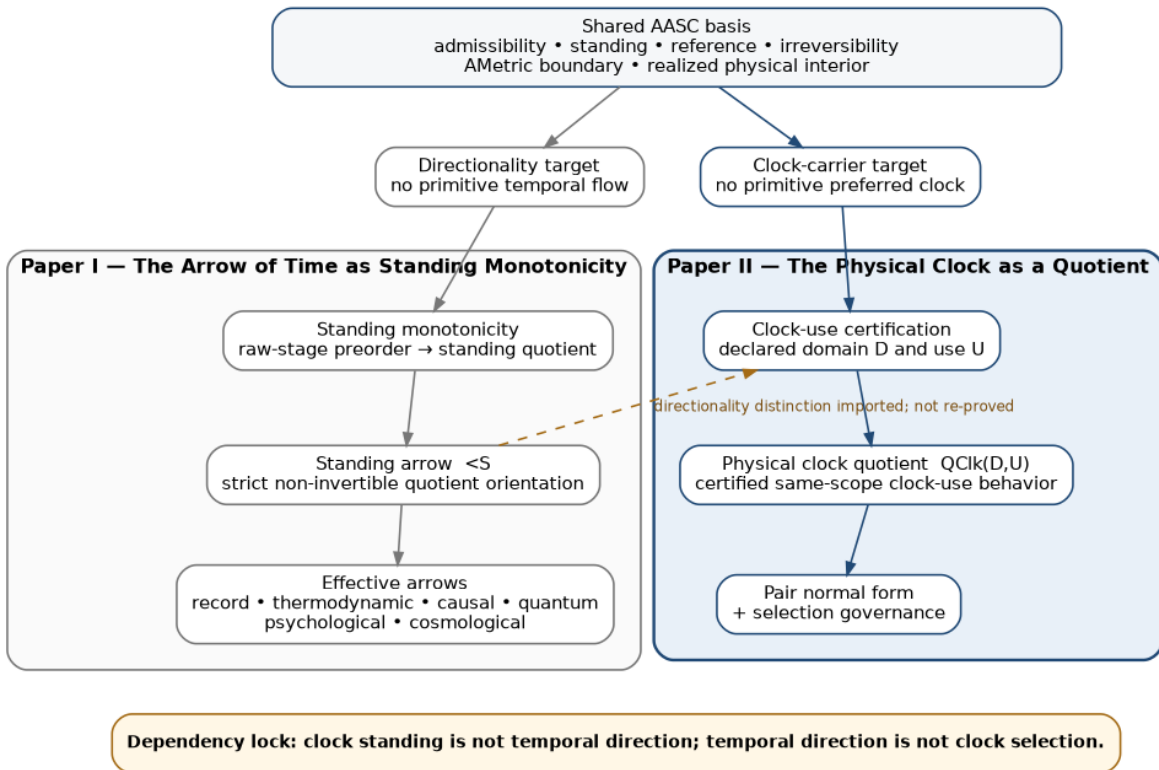


Figure 1: Architecture of the AASC time non-primitivity sequence, with Paper II emphasized. The present paper imports the directionality distinction from Paper I but does not re-prove or replace the standing arrow.

AASC constraint-formalism method

AASC is a constraint formalism. Its primary operational mode is exhaustion and impossibility rather than the proposal of a preferred clock, hidden temporal substrate, or substitute dynamics. Once the non-degenerate kernel and a fixed clock-use package are in force, AASC asks which candidate relations, selectors, representatives, repairs, transfers, performance criteria, or independent authorities remain available without changing the theorem object or breaking kernel faithfulness.

The contribution is closure over clock constructions, not another clock construction. The nontrivial result is the behavioral exhaustion that places every standing-relevant same-scope clock difference in evaluated classification or coherent retained continuation. The canonical quotient is the survivor form of that exhaustion; pair normal form and selection governance then close every residual relation and authority route.

This exhaustion-first posture is the source of the formalism’s novelty and force. A responsive objection must deny target instantiation, break a load-bearing upstream theorem, identify an invalid clock-specific specialization, or exhibit a kernel-faithful same-scope functional, relation, or selector outside the stated normal forms. A request for a preferred clock or a more familiar mechanism is not a refutation unless it supplies such a survivor.

Standing of objections. A criticism has standing only if it preserves the fixed clock-use target and kernel-faithful comparison class and then identifies an invalid specialization, an incomplete exhaustion branch, or a same-scope functional, relation, or selector outside the normal forms. Requests for a preferred variable, a familiar temporal mechanism, or a softer selection rule do not challenge the theorem unless they supply one of those witnesses.

Part I

Boundary, Scope, and Dependency Discipline

1 Introduction: internal time as a quotient problem

The target is clock-choice exhaustion. For a fixed constrained domain and declared use, AASC does not ask which representative variable is “really time.” It asks which candidates are certified, which differences are physical, which are anchor or theorem-object changes, and which attempts at singular preference possess declared authority. Once those routes are exhausted, the physical clock object is the final use-relative quotient and no residual same-scope category remains.

1.1 The clock-certification gap

The problem of time is usually introduced as a tension between ordinary quantum mechanics, which uses an external parameter in the Schrödinger equation, and general relativity, in which temporal and spatial geometry are dynamical. Canonical gravity is constrained; lapse and shift act as multipliers; the Hamiltonian constraint participates in refoliation or gauge-related structure; and physical content is sought on a reduced or otherwise constraint-defined state object. Quantization then yields a stationary constraint equation, schematically

$$\hat{\mathcal{H}}\Psi = 0, \tag{1}$$

whose interpretation depends on the constraint implementation, boundary conditions, physical inner product, and observable construction [DeWitt, 1967, Kuchař, 1992, Isham, 1993, Anderson, 2012, Kiefer, 2012].

Many established approaches answer the formal question by constructing relational observables, selecting matter clocks, conditioning on clock subsystems, deriving semiclassical parameters, or organizing records. The additional certification question is often left distributed across model-specific assumptions:

When does a formal temporal construction have standing as a reusable physical clock for a declared use?

A coordinate can order a derivation without carrying physical clock content. A monotonically varying phase-space function can fail to define a global gauge fixing. A conditional quantum state can depend on a clock POVM or physical inner product not fixed by the bare kinematic state. A record sequence can be ordered without fixing a clock role. This certification gap is the theorem target. AASC closes it by fixed-domain exhaustion; rival temporal-mechanism construction belongs to a distinct theorem class.

Table 1: From temporal construction to clock certification.

Question	Standard construction target	Additional clock-standing audit
What replaces external time?	Internal clocks, relational observables, records, semiclassical parameters	Which candidate is certified for the declared use?
Why do clocks disagree?	Multiple-choice problem	Are the candidates equivalent, eligibility-, anchor-, profile-, or theorem-object divergent, and is any singular choice lawfully governed?
When does time emerge?	WKB/background or conditional regimes	Which anchors, approximation orders, and failure boundaries are declared?
Are records enough?	Histories, decoherence, and record correlations	Are the records role-fixed, reusable, quotient-stable, and coherent?

Closure target

The closure target replaces the untyped question of which representative variable is “really time.” It asks whether same-scope clock choice has any residual form once the physical clock-use package is fixed. The capstone answer is negative: certified candidates are equivalent for the use, or their difference is anchor-level, quotient-profile-level, theorem-object-changing, selector-dependent, or fail-closed.

1.2 Established background and theorem contribution

Established background includes the following results: canonical quantum gravity does not inherit a unique external Schrödinger parameter; physical predictions can be formulated relationally; internal clocks may be local, periodic, nonideal, or globally obstructed; different clock choices can lead to inequivalent reduced descriptions or quantum theories; and Born–Oppenheimer/WKB time is tied to an approximation and a background split [Kuchař, 1992, Isham, 1993, Rovelli, 2002, Dittrich, 2006, Page and Wootters, 1983, Höhn et al., 2021, Kiefer and Singh, 1991, Gielen and Menéndez-Pidal, 2020].

The theorem contribution is final quotient closure. For a declared physical clock use, the manuscript derives the complete admitted probe family from the AMetric constructional-behavior exhaustion theorem rather than presuming any selected finite profile is complete. Every admissible clock-use functional then factors through the induced quotient. Candidate pairs are theorem-object divergent, eligibility divergent, anchor divergent, quotient-profile divergent, or physically equivalent. Singular selection is not inserted into that pairwise partition: it is a separate governance act exhausted by declared role fixation, prior boundary-trace fixation, scope enrichment, representative privilege, unsupported class privilege, and the no-generator/no-carrier/no-repair/no-bridge closure.

1.3 Correctness burden and dependency posture

This manuscript is downstream of stated fixed-domain kernel, exhaustion, quotient-finality, and impossibility theorems. The provenance of the formalism is not a premise of the proof. Because the target is a kernel-faithful clock-use package, an objection must preserve that package or explicitly deny target instantiation. A successful objection must do at least one of the following:

- (C1) deny that the declared clock-use domain instantiates the non-degenerate, kernel-faithful target class;
- (C2) break the kernel theorem or an imported exhaustion, quotient-finality, or impossibility theorem used by the local derivation;
- (C3) identify an invalid specialization from kernel faithfulness or another imported result to the clock target;
- (C4) exhibit an admissible clock-use functional that preserves the declared package but does not factor through the canonical clock quotient; or
- (C5) exhibit a same-scope clock relation or selection authority that preserves the declared comparison class while escaping the pair and governance normal forms.

Only those correctness routes bear on the argument. The capstone survivor burden is therefore precise: exhibit a residual same-scope clock identity, physical relation, or selector authority not determined by the canonical clock quotient and declared anchor structure.

Correctness burden. Deny target instantiation, break the kernel, identify a dependency error, or exhibit the survivor.

A criticism lacking one of those witnesses is a request for another theorem class, not a same-scope challenge to the present closure result.

1.4 Standard constructions and the added classification

The principal strategies in the literature supply distinct formal constructions and physical implementations. Partial and complete observables seek gauge-invariant correlations [Rovelli, 2002, Dittrich, 2006]; reduced and deparametrized quantization solve constraints relative to a selected clock [Thiemann, 2006]; material reference systems add physical clock and coordinate fields [Kuchař and Torre, 1991, Brown and Kuchař, 1995]; Page–Wootters and temporal-reference-frame programs recover relational quantum dynamics from constrained states and clock observables [Page and Wootters, 1983, Höhn et al., 2021]; thermal time uses modular flow of a declared algebra–state pair [Connes and Rovelli, 1994]; histories and records programs seek temporal organization in correlations and retained records; and semiclassical schemes recover approximate matter evolution relative to gravitational WKB structure [Kiefer and Singh, 1991, Kiefer and Wichmann, 2018].

AASC operates at the fixed-domain constraint layer common to all these routes. It determines which parts of a proposed clock construction are scope anchors, which are quotient-level physical content, which are representational surplus, and where clock eligibility fails.

Table 2: What established approaches supply and what the AASC audit adds.

Approach	Native contribution	Additional audit question
Dirac/complete observables	Constraint-invariant relational quantities	Does the chosen clock/readout descend and remain reusable for use U ?
Reduced or deparametrized quantization	A true Hamiltonian relative to a selected clock	Is the clock an anchor, a quotient-profile/tensor divergence, a theorem-object change, or an unlicensed selector?
Page–Wootters/reference frames	Conditional dynamics from a constrained state	Are the physical inner product, clock POVM, normalization, and failure profile certified?
Material reference systems	Physical clock and coordinate fields	Is the matter already in scope, or is the clock a declared enrichment?
WKB/semiclassical time	Effective evolution relative to a background branch	Which split, branch, gauge, order, and breakdown conditions define the effective clock?
Thermal time	Modular flow of an algebra–state pair	What operational readout turns the modular parameter into a physical clock?
Histories and records	Temporal organization without primitive external time	Are records role-fixed and reused in one quotient-stable clock lineage?

Relational time survives as admitted physical content once its standing conditions are certified. A relational observable can be formally constructed without settling whether its clock variable is eligible for a physical use, whether rival clocks are canonically equivalent or quotient-profile/tensor-divergent, whether a record profile is reusable, or whether failure is local and fail-closed. The present manuscript closes those questions on the declared comparison class.

The classification is the theorem: it exhausts the admissible forms of clock disagreement. A candidate can fail by lacking descent, by lacking an adequate readout, by crossing its validity boundary, by changing the theorem object, or by importing an undeclared selector. These diagnoses have different consequences and cannot be collapsed into one generic “clock ambiguity.”

1.5 Local normal form induced by the kernel roles

The following formulation isolates the local inference after the kernel roles have been imported. It is not a weakening of the AASC dependency. Consider any constrained theory with:

- (A1) a fixed physical equivalence or redundancy relation;
- (A2) physical observables invariant under that relation;
- (A3) a specified practice in which clock readouts are reused to condition or order physical content;
- (A4) an irreversible distinction between successful and failed clock use on the retained history; and
- (A5) no undeclared selector among physically equivalent clock candidates.

Then the admitted clock-use behavior has a canonical probe quotient: candidates are equivalent exactly when every licensed clock-use probe agrees on definedness and value. The same fixed-domain roles then force a physical pair normal form and a separate selection-governance normal form. AASC supplies the necessity, behavior-exhaustion, quotient-finality, and impossibility results used to prove that no further same-scope functional or selector survives.

1.6 Target burden and use relativity

Let U denote the physical task for which a candidate is submitted. Examples include ordering records, constructing a complete observable, generating a reduced evolution, synchronizing subsystems, or recovering proper time. Clock standing is written

$$\text{ClockStand}_D^U(\Theta), \quad (2)$$

and is always relative to both D and U . A candidate can be adequate for one use and inadequate for another. Bivalence applies to eligibility for the specified use; resolution, accuracy, drift, and backreaction may remain graded inside an admitted clock model.

Clock-standing claim at a glance

Every reportable clock-standing claim identifies: (i) the declared use; (ii) the active domain and boundary conditions; (iii) the candidate readout; (iv) its constraint and physical-descent certificate; (v) the record or conditioned-use ledger; (vi) the continuation and failure rules; and (vii) the relation to rival clocks. The full ten-coordinate certificate is given in [section 7](#).

1.7 Terminology lock

“Standing” means eligibility for licensed reuse in the declared target. It does not mean existence, truth, probability, empirical confirmation, or ontological fundamentality. “Tensor” in the AASC role taxonomy means quotient-level invariant load; it need not be a tensor field or a multilinear tensor in the ordinary mathematical sense. Reader-facing references to *quotient-profile/tensor divergence* name this invariant-profile branch. “Redescription” means transformations declared physically redundant by the target theory, not every transformation called gauge in every interpretation.

1.8 Contributions and theorem hierarchy

The manuscript’s theorem hierarchy is:

1. the *Clock-Standing Certificate Criterion*;
2. the *No Clock-to-Kernel or Clock-to-Arrow Backflow Theorem*;
3. the *Clock-Profile Sufficiency and Necessity Theorem*;
4. the *Clock-Profile Finality Theorem*;
5. the *Clock-Quotient Uniqueness Theorem*;
6. the *Clock-Pair Physical Normal-Form Theorem*;
7. the ordered *Clock-Selection Governance Theorem*;
8. the capstone *Physical-Clock Quotient Finality and Choice-Closure Theorem*;

9. the quotient-status and saturated-registry completion corollaries; and
10. derived classifications of same-scope external-clock import, frozen formalism, local clock failure, semiclassical time, ADM presentation variables, material clocks, Page–Wootters constructions, thermal time, histories, and quantum-cosmology clock dependence.

The central endpoint is not a selected preferred clock. It is the final canonical quotient of physical clock-use behavior and the closure of every pair relation and selection act around that quotient.

1.9 Constraint-formalism method: exhaustion and impossibility

The proof method is fixed-domain exhaustion followed by quotient finality. The clock comparison class is locked first. AMetric constructional-behavior exhaustion then proves that every standing-relevant clock-use distinction enters through evaluated classification or coherent retained continuation. ATS and AER force profile sufficiency, profile necessity, quotient factorization, and no residual refinement. The apex closure theorem supplies no competing same-domain gate or admissible interior, while the Impossibility Suite closes attempted generation, transfer, accumulation, repair, or permissive bridge routes. The pair and selection normal forms therefore follow as formal consequences of the final quotient and its governance envelope.

These are fixed-domain constraint theorems. Their force lies in route exhaustion rather than mechanism construction: once the package remains kernel-faithful, a purported new clock functional, pair relation, or selector must either enter the declared behavior loci, change scope, import independent authority, collapse to skin, or fail. The quotient universal property is applied only after that closure has been established.

The refutation burden follows the theorem class. A critic must deny target instantiation, break an imported closure theorem, identify an invalid specialization, or exhibit a faithful same-scope survivor. Source familiarity and requests for a preferred clock have no bearing unless they produce that survivor.

2 Imported closure spine and dependency posture

This manuscript is downstream of the AASC primitive, fixed-domain, and physics theorem objects. The relevant audit is whether the target clock domain instantiates their hypotheses, whether the imported statements are correct, and whether the clock-specific inference follows. Source popularity, familiarity, or author-origin has no evidential role.

2.1 AMetric fixed-domain closure spine

The AMetric boundary paper supplies the exact local closure architecture needed by the clock theorem [Maley, 2026b]:

- (A1) Definition 2.1 and Theorem 2.2 lock the comparison class by boundary typing, evaluated fragment, standing classification, lawful quotient, retained continuation, and anchor–tensor–skin role carrier.
- (A2) Theorem 2.5 excludes selector, parameter, metric, probability, typicality, index, canonical representative, and second-classifier authority at the boundary.
- (A3) Theorems 2.8 and 2.10 exhaust constructional behavior by evaluated-act classification and coherent retained-trajectory structure.

- (A4) Theorems 2.12 and 2.13 exhaust and collapse selector import, parameter injection, deferred admissibility, repair, second classifier, scope transport, representative privilege, and tensor/mechanism reification.
- (A5) Theorem 2.15 and Corollary 2.16 make anchor, tensor, and skin exhaustive and mutually role-determining on the fixed domain.
- (A6) Theorem 14.2 exports the same boundary closure to every downstream regime that independently instantiates the admissibility-bearing target class.

The clock-specific probe family, pair normal form, and selection-governance theorem below are direct specializations of this closure spine after the clock package has been fixed.

2.2 Apex closure and uniqueness of the admissible interior

Unus Solus Possibilis Est supplies the same-domain apex closure used to block competing clock-governance regimes [Maley, 2026r]. Its Lemma 4 excludes disagreeing same-domain gates; Theorems 1–3 exclude repair, generation, and carrier authorization; Theorem 4 excludes cross-domain standing transfer without target gating; Lemma 7 and Theorem 6 eliminate discrete multiplicity of admissible interiors; and Theorem 7 packages the closed regime.

This import does *not* imply that a fixed physical interior contains only one clock class. Interior uniqueness concerns the standing gate and admissible construction regime. A plural clock quotient is legitimate when its classes differ by declared anchors or quotient-profile/tensor content inside that one interior. What cannot survive is a second same-domain clock authority or admissibility regime introduced to select among them.

2.3 Predecessor arrow theorem and series dependency

The immediate predecessor in the time sequence is *The Arrow of Time as Standing Monotonicity* [Maley, 2026d]. Its role here is narrow but essential: it blocks the conflation of clock standing with temporal direction. The physical clock quotient does not ground the standing arrow; it certifies clock-use carriers inside already admitted, standing-governed scopes. Conversely, the arrow theorem does not select a preferred clock. This one-way dependency preserves the series order: Paper I closes directionality; Paper II specializes clock identity, plurality, failure, and selection.

2.4 Imported theorem ledger

The table fixes the remaining imported objects and their local uses. The accompanying source manifest records the cited files, version dates, theorem identifiers, and SHA-256 hashes.

Table 3: Imported theorem ledger.

Source	Imported result	Local use
Kernel paper	Non-degenerate construction forces admissibility, standing, reference, and irreversibility; no faithful lower generator or strict same-domain weakening	Establishes that clock-use certification begins only after target instantiation and kernel faithfulness. The clock theorem does not derive the kernel; it proves the clock-specific quotient consequence [Maley, 2026l].
Structure of admissibility	AMetric interface, bivalence, prior gate, and standing conservation	Excludes graded clock standing and a second positive-side clock classifier [Maley, 2026p].

Source	Imported result	Local use
AMetric boundary closure	Comparison-class lock; no selector/parameter; constructional-behavior exhaustion; route exhaustion/collapse; ATS role discipline; regime-invariant export	Supplies the selection, pairwise, and probe-family closure spine [Maley, 2026b].
Unus apex closure	Gate uniqueness; no repairs/generators/carriers; no standing transfer; no discrete admissible interiors; apex closure	Blocks a second same-domain clock-governance regime while preserving typed clock plurality within one interior [Maley, 2026r].
Trajectory theory	Prefix/composition closure, irrecoverability, and downstream failure	Supplies no deferred clock coherence and fail-closed clock histories [Maley, 2026q].
ATS decomposition	Anchor/tensor/skin decomposition, quotient factorization, finality, and no residual refinement	Supplies the role partition and final standing quotient [Maley, 2026c].
Fixed-domain exhaustion	Standing-presentation separation and no same-scope consequence outside fixed behavior loci	Supports exhaustiveness of the admitted probe family [Maley, 2026i].
Constraint architectures	Evaluation coherence and constructional-behavior exhaustion	Blocks licensed distinctions outside the fixed witness/probe basis [Maley, 2026f].
Impossibility Suite	No generators, carriers, repairs, accumulation, or permissive bridge principles	Closes success, utility, probability, explanation, repetition, transfer, and repair as clock-authority sources [Maley, 2026k].
Boundary-trace fixation	Conditional singleton trace fixation and converse/no-go	Gives a lawful fixation route only when the trace is already declared or independently admitted before selection [Maley, 2026e].
Continuation/persistence	Continuation-locus induction and persistence realization	Types physical clocks as continuation-bearing roles rather than bare variables [Maley, 2026g,n].
AER	Characteristic identification, first-class closure, multiplier form, and GR/QM quotient architecture	Fixes the constraint quotient, physical observables, and presentation/redescription structure [Maley, 2026a].
Gravity paper	AASC gravity role	Locates the clock problem inside the constrained gravitational interior [Maley, 2026j].
Einstein projection	Minimal first-response metric projection	Keeps metric and proper-time clocks downstream of licensed metric structure [Maley, 2026h].
Metricity licensing	L-metric licensing and no metric backflow	Blocks metric time from authorizing the domain in which it is defined [Maley, 2026m].
Record paper	Measurement-record quotient and fixation-commutation results	Supplies the base record-use burden; clock ordering/conditioning is added here [Maley, 2026o].
Arrow paper / Time Non-Primitivity I	Standing monotonicity; distinction among the standing arrow, physical time, records, and effective arrows	Supplies the predecessor directionality boundary and prevents clock standing or clock selection from being conflated with temporal direction [Maley, 2026d].

2.5 Exact clock-theorem dependency map

The following map fixes the imported closure object used at each load-bearing clock-specific step. It is a dependency map, not an additional proof layer.

Table 4: Exact dependency map for the quotient-finality theorem package.

Clock-specific result	Imported closure object
Clock comparison-class lock	AMetric Definition 2.1 and Theorem 2.2: preservation of boundary typing, evaluated fragment, standing classification, lawful quotient, retained continuation, and role carrier.
Clock/arrow separation	Paper I standing monotonicity and effective-arrow taxonomy: clock standing is a downstream carrier status, not temporal direction or an arrow source.
Clock-use behavioral exhaustion	AMetric Theorems 2.8 and 2.10 and Corollary 2.9: every standing-relevant effect enters evaluated-act classification or coherent retained-trajectory structure.
Complete clock-use probe family	AMetric behavioral exhaustion together with ATS standing-identity finality/no residual refinement and AER fixed-domain physical exhaustion.
Profile sufficiency and necessity	AMetric two-locus exhaustion, ATS standing-presentation separation, and AER quotient/redescription closure.
Clock-profile finality	ATS finality and quotient factorization, AMetric no-third-locus closure, AER no residual same-scope refinement, and <i>Unus</i> no competing gate/carrier/transfer regime.
Clock-quotient uniqueness	ATS standing-quotient uniqueness and the universal property of the canonical probe quotient.
Clock-pair physical normal form	AMetric comparison-class lock and ATS anchor-tensor-skin role exhaustion, after quotient finality.
Clock-selection governance	AMetric no-selector/no-parameter theorem and route exhaustion/collapse; boundary-trace fixation for the prior singleton-trace route; <i>Unus</i> gate/interior uniqueness; Impossibility-Suite no-generator/no-carrier/no-repair/no-bridge closure.
Saturated completion	<i>Unus</i> domain individuation and scope discipline together with the universal clock-choice closure theorem.

2.6 General quotient form

The imported AER paper proves a regular-stratum Hamiltonian result using a smooth quotient map. The present paper does not assume that every quantum-gravity realization has a global smooth reduced space. It therefore uses an abstract physical quotient object $\mathbf{Q}_D$. A clock-conditioned construction descends when it is invariant under the declared redundancy relation and defines an element of $\mathbf{Q}_D$: a reduced observable, physical operator class, group-averaged/rigged observable, physical-history class, or another explicitly declared physical object. When a smooth quotient exists, descent reduces to

$$F = \pi^* \tilde{F} = \tilde{F} \circ \pi. \quad (3)$$

This distinction prevents the classical regular-stratum notation from being silently globalized to singular, algebraic, or non-reduced quantum domains.

2.7 Source provenance

The accompanying `SOURCE_MANIFEST.md` records the cited AASC source files, version dates, imported theorem identifiers, and SHA-256 hashes. The local clock results are conditional on those imported theorem objects. Their correctness depends on the imported statements and the validity of the present inferences, not on any external status classification.

2.8 Non-circularity

The manuscript assumes no preferred clock, deparametrization, material reference system, Page–Wootters subsystem, semiclassical branch, thermal state, or global time function. It assumes the fixed constraint/redundancy object and the stated clock-use burden. The local work is to derive the divergence partition, ordered normal form, and use-relative necessity criteria. No clock result is used to prove the upstream AASC or AER theorems.

3 The problem of time as a fixed-domain target

3.1 Canonical origin and scope guard

In a parametrized or generally covariant system the canonical Hamiltonian is constrained. For canonical general relativity one writes schematically

$$H_{\text{tot}}[N, N^i] = \int_{\Sigma} d^3x \left(N\mathcal{H} + N^i\mathcal{H}_i \right), \quad (4)$$

where N and N^i are lapse and shift. On a compact hypersurface without boundary, or with boundary data and charges held fixed, bulk changes of these multipliers alter representative developments rather than the reduced physical state. With nontrivial boundaries, asymptotic symmetries, or physical boundary Hamiltonians, the corresponding boundary data can instead be anchor or tensor content and are not classified as skin by default [Dirac, 1964, Henneaux and Teitelboim, 1992, Regge and Teitelboim, 1974, Maley, 2026a].

At the quantum level the constraint equations may be implemented by reduced quantization, Dirac quantization, group averaging, algebraic methods, path integrals, or other physical-state constructions. The paper does not assume that all of these produce the same global mathematical quotient. It fixes only the physical equivalence, observable, record, and continuation structures declared by the target theory.

3.2 Problem-of-time facets

Table 5: Problem-of-time facets and the present standing question.

Facet	Standard form	Fixed-domain clock question
Frozen formalism	Constraint equation lacks external Schrödinger time	Is external evolution absent, or merely not represented on the physical object?
Multiple choice	Different internal clocks can yield inequivalent descriptions	Where is the first divergence: theorem object, eligibility, definedness, or physical profile?
Observables	Physical quantities must respect the constraints	Does the conditioned construction define an element of $\mathbf{Q}_D$?
Global time	A clock can have turning points or fail globally	What is its declared patch and failure certificate?
Foliation dependence	Slicings alter representative development	Are boundary data fixed, and is the foliation skin, anchor, or tensor content?
Semiclassical time	Time appears in WKB/Born–Oppenheimer sectors	Which split, branch, gauge, and error order constitute the effective package?
Records/histories	Change is reconstructed from correlations or records	Are readouts fixed, reusable, and coherent for the stated use?
Inner-product dependence	Clock choice can affect physical Hilbert structure	Is the comparison same-scope, or has the theorem object changed?

3.3 Fixed constrained clock domain

Definition 3.1 (Constrained quantum-gravity clock domain). A fixed constrained clock domain is

$$\mathcal{D}_{\text{Clk}} = (\mathcal{K}, \mathcal{C}, \sim_D, \mathbf{Q}_D, \mathcal{O}_{\text{phys}}, \mathcal{R}, \mathcal{T}, \text{Cont}_D, \text{Adm}_D), \quad (5)$$

where $\mathcal{K}$ is a kinematic or history carrier; $\mathcal{C}$ is the physical constraint/solution condition; $\sim_D$ is the declared redundancy relation; $\mathbf{Q}_D$ is the physical quotient object; $\mathcal{O}_{\text{phys}}$ is the physical observable family; $\mathcal{R}$ is the record family; $\mathcal{T}$ is the clock-candidate class; Cont_D is retained continuation; and Adm_D is the use-eligibility predicate.

The tuple is a comparison schema that leaves the native formalism intact. In a regular classical presentation $\mathbf{Q}_D$ may be a smooth reduced phase space with projection $\pi : C \rightarrow S_{\text{phys}}$. In a quantum presentation it may instead be a physical Hilbert space, rigged/group-averaged state space, operator algebra modulo constraints, or a physical histories object. Every theorem below is relative to the declared instantiation.

Definition 3.2 (Clock-use target). A use U specifies what the candidate is required to do: construct a relational observable, order or condition records, generate reduced evolution, synchronize subsystems, or represent proper time. A candidate realizes the target only when its output is intended to survive lawful redescription and later same-domain reuse for that use.

Definition 3.3 (Same-scope clock presentation). Two candidates are compared on the same scope only when they preserve, up to lawful equivalence: the constraint/solution object; the redundancy relation; Q_D ; the physical observable and record families; the continuation and failure rules; boundary charges and declared boundary Hamiltonians; and the use U . Altering any of these is theorem-object change rather than a same-scope refinement.

Remark 3.4 (Regularity boundary). Global singular quotients, Gribov regions, non-Hausdorff orbit spaces, changing constraint rank, and inequivalent physical inner products require separate treatment. The present normal form still classifies the role of added structures, but no smooth global quotient or global clock is inferred.

4 Clock-role taxonomy and terminal dispositions

4.1 Same-domain roles

The imported ATS theorem partitions same-domain descriptive work into three roles. These are not three kinds of substance.

Definition 4.1 (Clock anchor). A clock anchor is a declared precondition whose variation changes whether the use U is well typed. Examples include a background or semiclassical geometry, a material reference field already present in the theory, a clock subsystem and POVM, a boundary trace, an orientation/positive-frequency sector, a physical inner product, or a local monotonicity patch.

Definition 4.2 (Clock tensor content). Clock tensor content is quotient-level relational or record content whose variation changes the physical clock profile while anchors are fixed. Examples include complete observables, physical conditional states, quotient-stable readouts, clock-conditioned transition amplitudes, and certified overlap maps between local clocks.

Definition 4.3 (Clock skin). Clock skin is representative surplus whose variation changes neither use definedness nor the physical profile. Coordinate labels, gauge representatives, basis choices, parametrizations, and bulk lapse/shift choices are skin only under this invariance condition.

4.2 Terminal meta-dispositions

Richer scope and fail-closed are not fourth and fifth ATS roles. They are terminal dispositions applied before same-domain role typing.

Definition 4.4 (Richer-scope disposition). A candidate is richer scope relative to D when it changes the state carrier, constraints, physical quotient object, observable algebra, physical inner product, boundary Hamiltonian or charges, record-use burden, or continuation rules. The added structure may be physically legitimate; it is not a same-scope clock derivation.

Definition 4.5 (Fail-closed disposition). A same-scope candidate is fail-closed for U when it lacks quotient descent, a valid readout for that use, coherent continuation, a declared validity domain, or selector-free role fixation. Failure is use-relative and does not imply that the variable is mathematically meaningless.

4.3 Role ledger

Table 6: Initial role ledger for clock-related structures.

Candidate	Default classification	Qualification
Coordinate label t	Skin	Anchor only in an explicitly background-dependent theory
Bulk lapse/shift	Skin/presentation	Boundary values associated with physical charges or Hamiltonians are anchor/tensor, not skin
Foliation	Skin, anchor, or richer scope	Depends on whether it changes physical content or boundary data
Proper time	Metric-interior tensor content	Requires a licensed worldline, metric regime, calibration, and readout
Scalar field ϕ	Candidate anchor/tensor	Requires a local nondegenerate gauge-fixing patch and physical readout
Dust/reference matter	Relative-scope dependent	Richer scope relative to vacuum gravity; possibly same-scope if already in the base theory
Page–Wootters subsystem	Candidate anchor/tensor	Requires physical constraint, inner product, clock POVM, and conditional normalization
Modular flow	Package-relative quotient-profile/tensor candidate	Requires a suitable algebra, faithful state/weight, and operational readout map
Record sequence	Trace or quotient-profile/tensor candidate	Becomes clock content only after readout-role fixation and coherent reuse
Semiclassical WKB time	Effective anchor/tensor	Valid only for the declared split, branch, gauge, and approximation order
Probability/typicality	Interior tensor or skin	Never an undeclared selector of clock standing

Remark 4.6 (Probability and quality). Born weights, probabilities, accuracy scores, and typicality measures may be legitimate physical content inside an admitted clock model. They are excluded only when used as undeclared authority to decide which physically equivalent candidate occupies the clock role.

Part II

Clock Domains, Records, and Quotient Structure

5 Formal clock candidates, admitted probe behavior, and the canonical clock quotient

5.1 Clock-use package

Definition 5.1 (Clock-use package). A constrained physical clock-use package is

$$\mathfrak{C}_D^U = (D, U, Q_D, \text{Read}_D, \text{Rec}_D, \text{Cont}_D, \text{Fail}_D), \quad (6)$$

where D is the fixed constrained domain, U is the physical task, Q_D is the physical quotient object, Read_D and Rec_D fix the readout and reuse burdens, Cont_D fixes retained continuation, and Fail_D fixes the failure boundary.

The package fixes the theorem object against which clock standing, equivalence, divergence, selection, realization, and plurality are evaluated. None of those statuses is absolute. A candidate may stand for one use and fail for another.

A clock-use package is **kernel-faithful** when its evaluated candidates preserve admissibility, standing, reference, irreversibility, no same-act repair, lawful redescription, the physical quotient object Q_D , the declared readout/record burden, retained continuation, and the failure boundary. A package that introduces a clock, probability, typicality rule, representative, success criterion, external parameter, or boundary trace as standing-generating authority changes scope unless that structure was already declared as anchor or quotient-profile content before the selection act.

Definition 5.2 (Clock comparison class). A clock construction remains same-scope for $\mathfrak{C}_D^U$ precisely when it preserves simultaneously:

- (C1) the clock-use boundary signature and admissibility-typing discipline;
- (C2) the evaluated clock-use fragment;
- (C3) standing/admissibility classification of evaluated clock-use acts;
- (C4) the lawful redescription relation and physical quotient object Q_D ;
- (C5) the retained finite-history readout, record, failure, and continuation structure; and
- (C6) the role carrier on which anchor, quotient-profile/tensor, and skin assignments are evaluated.

Proposition 5.3 (Clock comparison-class lock). *A proposed clock variation that alters any component of theorem 5.2 is not a same-scope refinement. It is theorem-object divergence or declared scope enrichment. A faithful same-scope challenge must preserve all six components while carrying a standing-relevant difference.*

Proof. The six components individuate the clock-use theorem object: boundary typing, evaluated material, gate classification, quotient identity, retained continuation, and role discipline. Altering one changes what is evaluated. Preserving all six while changing neither evaluated classification nor retained continuation changes no constructional behavior and is skin by AMetric Theorems 2.8–2.10 [Maley, 2026b]. $\square$

5.2 Formal and physical candidates

Definition 5.4 (Formal relational clock candidate). A formal relational clock candidate is

$$\Theta_{\text{form}} = (T, D_T, \mathcal{O}_T), \quad (7)$$

where T is a variable, subsystem, field, flow, or label; D_T is its formal domain; and $\mathcal{O}_T$ is a family of conditional, relational, partial, or complete observables constructed relative to T .

Definition 5.5 (Physical clock-standing presentation). For a declared use $\mathbf{U}$, a physical presentation is

$$\Theta_{\text{stand}}^{\mathbf{U}} = (\Theta_{\text{form}}, \text{Read}_T, \text{Rec}_T, \text{Cont}_T, \text{Fail}_T), \quad (8)$$

where Read_T fixes the readout relation, Rec_T is the reusable record ledger required by $\mathbf{U}$, Cont_T is retained continuation, and Fail_T states the validity and failure domain.

Let $\mathcal{C}_D^{\mathbf{U}}$ be the full class of evaluated background-same-scope clock candidates submitted to the package, and let

$$\text{Cert}_D^{\mathbf{U}} = \{\Theta \in \mathcal{C}_D^{\mathbf{U}} : \text{ClockStand}_D^{\mathbf{U}}(\Theta)\} \quad (9)$$

be the certified class. A bare parameter may order a representative history or equation without defining an element of $\mathbf{Q}_D$ or a fixed readout relation for $\mathbf{U}$; such a parameter need not belong to $\text{Cert}_D^{\mathbf{U}}$.

5.3 Complete admitted clock-use probes

Lemma 5.6 (Clock-use behavioral exhaustion). *Inside the fixed clock comparison class, every standing-relevant clock-use distinction alters at least one of two loci:*

- (B1) *evaluated clock-use classification, including eligibility and definedness; or*
- (B2) *coherent retained clock-record and continuation structure, including quotient output, readout, order, performance-relevant behavior, and failure propagation.*

A distinction changing neither locus is skin. A distinction obtaining force only by changing the comparison class is theorem-object divergence or scope enrichment. No third same-scope clock-use locus remains.

Proof. This is the clock specialization of AMetric constructional-behavior exhaustion, Theorems 2.8 and 2.10 [Maley, 2026b]. Clock admission, rejection, readout-definedness, and failure classification occupy the evaluated-act locus. Record reuse, ordering, conditioning, performance with use consequences, and downstream continuation occupy the retained-history locus. A proposed third locus either changes one of these or changes neither; in the latter case it affects no standing-relevant clock use. ATS and AER supply the corresponding anchor/profile/skin and quotient/redescription specializations [Maley, 2026c,a]. If a purported third locus instead installs a distinct same-domain classifier, gate, or admissible interior, it is excluded by the no-competing-gate and no-discrete-interior results of *Unus* [Maley, 2026r]. Thus neither a hidden behavior locus nor a second governance locus survives. $\square$

Definition 5.7 (Complete admitted probe family). The complete admitted clock-use probe family $\mathcal{H}_D^{\mathbf{U}}$ is the class of every licensed same-scope probe whose result can affect the declared clock use. Its probes test, where applicable:

- (H1) theorem-object preservation and use eligibility;

- (H2) clock-use definedness and anchor conditions;
- (H3) quotient-descending relational or conditioned observables;
- (H4) readout identity, record reuse, and record order;
- (H5) coherent retained continuation;
- (H6) performance data when those data change the admitted use; and
- (H7) declared failure behavior.

The family is extensionally complete relative to the clock comparison class: an alleged additional same-scope functional with standing-relevant clock-use effect changes one of the two loci in [theorem 5.6](#) and is therefore already an admitted probe. A functional changing neither locus is skin; one requiring a new probe, gate, quotient, state variable, boundary channel, or continuation rule changes scope.

The nontrivial claim is not that a quotient can be formed once a complete probe family is given. The nontrivial claim is that, relative to the fixed kernel-faithful comparison class, every standing-relevant clock-use difference must enter evaluated classification or coherent retained continuation. A proposed additional clock-use functional is responsive only if it preserves the same $(D, U, Q_D, \text{Read}_D, \text{Rec}_D, \text{Cont}_D, \text{Fail}_D)$ package, carries standing-relevant clock-use force, and changes neither locus. The clock-use behavioral-exhaustion lemma excludes that third locus. Only after this exhaustion result is the quotient universal property routine rather than stipulative.

The coverage claim follows from the imported exhaustion theorems rather than from an open-ended enumeration. It uses AMetric constructional-behavior exhaustion (Theorems 2.8 and 2.10), ATS standing-identity finality and no residual same-scope refinement, and AER fixed-domain physical exhaustion. The exact imported source objects are recorded in the accompanying dependency ledger and corpus-control extract: AMetric Definition 2.1 and Theorems 2.2, 2.8, and 2.10; the ATS finality, quotient-uniqueness, skin-non-authority, and no-residual-refinement results; and the AER bridge, unique-classifier, no-second-selector, and no-surviving-rival results. A proposed new clock functional defeats the theorem only by preserving the full clock comparison class, carrying standing-relevant force, and failing to factor through $\mathcal{H}_D^U$. Such a functional would have to alter neither evaluated classification nor retained continuation while nevertheless changing clock use, contradicting [theorem 5.6](#).

For $h \in \mathcal{H}_D^U$ and $\Theta \in \mathcal{C}_D^U$, write

$$\text{Pred}_D^U(h; \Theta) \downarrow \tag{10}$$

when the probe is defined, and let $\text{Pred}_D^U(h; \Theta)$ denote its value. Definedness is part of the behavior.

Definition 5.8 (Canonical clock equivalence). For certified candidates,

$$\Theta \sim_D^U \Phi \iff \forall h \in \mathcal{H}_D^U, \text{Pred}_D^U(h; \Theta) = \text{Pred}_D^U(h; \Phi) \tag{11}$$

in both definedness and value. The canonical physical clock quotient is

$$\mathcal{Q}_{\text{clk}}(D, U) := \text{Cert}_D^U / \sim_D^U, \tag{12}$$

with quotient map $q_D^U : \text{Cert}_D^U \rightarrow \mathcal{Q}_{\text{clk}}(D, U)$.

The quotient is defined by total admitted clock-use behavior, not by a preferred variable or an arbitrary finite list of features.

5.4 Finite presentation profile

Definition 5.9 (Clock anchor signature). The clock anchor signature $A_D^U(\Theta)$ records the same-domain conditions under which the use is defined: active sector, boundary data, gauge slice, monotonic or periodicity domain, approximation order, readout-definedness conditions, and any declared background or material-clock anchor.

Definition 5.10 (Invariant clock-use profile). The invariant physical profile is

$$P_D^U(\Theta) = \left(Q_\Theta^U, \text{Read}_\Theta^U, \text{Rec}_\Theta^U, \text{Ord}_\Theta^U, \text{Cont}_\Theta^U, \text{Perf}_\Theta^U, \text{Fail}_{\Theta, \text{int}}^U \right), \quad (13)$$

where Perf records accuracy, resolution, stochasticity, drift, decoherence, and backreaction only when they affect admitted physical use, and Fail_{int} records failure behavior after the anchor domain is fixed.

Define the displayed presentation

$$r_D^U(\Theta) = (A_D^U(\Theta), P_D^U(\Theta)). \quad (14)$$

The notation $\text{Prof}_D^U(\Theta)$ may be used as shorthand for this ordered pair.

Theorem 5.11 (Clock-Profile Sufficiency and Necessity Theorem). *The displayed profile is sufficient and necessary for canonical clock equivalence:*

$$r_D^U(\Theta) = r_D^U(\Phi) \implies \Theta \sim_D^U \Phi, \quad (15)$$

$$\Theta \sim_D^U \Phi \implies r_D^U(\Theta) = r_D^U(\Phi). \quad (16)$$

Together, profile sufficiency and profile necessity make r_D^U a sufficient-and-necessary presentation of the canonical quotient.

Proof. For profile sufficiency, every admitted clock-use probe changes either use eligibility/definedness or coherent retained clock-use behavior. The first locus is recorded by A_D^U ; the second is recorded by quotient output, readout, record, order, continuation, performance-relevant content, and interior failure in P_D^U . This is the clock specialization of AMetric Theorems 2.8 and 2.10, the ATS standing-presentation and no-refinement theorems, and the AER two-locus bridge theorem for quotient content and redescription/continuation structure [Maley, 2026b,c,a,f,i]. Equality of the displayed coordinates therefore forces agreement of every admitted probe.

For profile necessity, suppose $\Theta \sim_D^U \Phi$ but the displayed profile differs. A differing coordinate would either change an admitted clock-use probe, contradicting canonical equivalence, or distinguish candidates without any admissible consequence. The latter is representative privilege or residual same-scope refinement beneath tensor content and is skin by ATS finality [Maley, 2026c]. If the difference is instead claimed to govern eligibility or continuation through an independent same-domain classifier, it contradicts *Unus* gate uniqueness and no-discrete-interior closure [Maley, 2026r]. Hence the profile cannot differ. $\square$

Theorem 5.12 (Clock-Profile Finality Theorem). *Every admissible present or future same-scope clock-use functional factors uniquely through the canonical quotient, where “future” means generated by lawful continuation inside the fixed kernel-faithful $(D, U, \mathcal{H}_D^U)$ package rather than by adding a new probe, selector, state variable, boundary channel, or theorem-object component. The factorization*

conclusion is downstream of kernel faithfulness and does not generate admissibility, standing, reference, irreversibility, physical time, or temporal direction. Specifically, if

$$F : \text{Cert}_D^{\mathcal{U}} \rightarrow Y \quad (17)$$

is admissible and clock-use relevant in that fixed sense, then there exists a unique

$$\overline{F} : \mathcal{Q}_{\text{Clk}}(D, \mathcal{U}) \rightarrow Y \quad (18)$$

such that

$$F = \overline{F} \circ q_D^{\mathcal{U}}. \quad (19)$$

Proof. Let $\Theta \sim_D^{\mathcal{U}} \Phi$. If F is treated as generating clock standing rather than evaluating certified clock-use behavior, then it is not a same-scope clock-use functional; it is a proposed selector, carrier, generator, or scope enrichment and is governed by the later selection theorem. For a genuine same-scope functional, if $F(\Theta) \neq F(\Phi)$, then F changes either evaluated clock-use classification/definedness or coherent retained continuation. By AMetric constructional-behavior exhaustion, it is therefore already an admitted probe in $\mathcal{H}_D^{\mathcal{U}}$, contradicting canonical equivalence [Maley, 2026b]. If it changes neither locus, it is skin. If it acquires relevance only by adding a new probe, classifier, operator, state variable, boundary channel, or continuation rule, the comparison class changes. If it purports to remain same-domain by installing an independent gate, carrier, or admissible interior, it contradicts *Unus* gate uniqueness, no-carrier/no-transfer closure, and no-discrete-interior theorem [Maley, 2026r]. If it derives authority from later success, probability, utility, explanation, repetition, cross-domain inheritance, or repair, it collapses under no-generation, no-carrier, no-accumulation, no-repair, and no-bridge [Maley, 2026k]. Therefore F is constant on canonical classes, and the quotient universal property gives unique factorization. $\square$

Theorem 5.13 (Clock-Quotient Uniqueness Theorem). *Every quotient-respecting and separating same-scope presentation of physical clock content is uniquely isomorphic to $\mathcal{Q}_{\text{Clk}}(D, \mathcal{U})$. Here quotient-respecting means canonical-equivalent candidates receive the same presented value, and separating means equal presented values imply canonical equivalence. Consequently, the standing clock object is the canonical quotient class $[\Theta]_D^{\mathcal{U}} = q_D^{\mathcal{U}}(\Theta)$, not a privileged representative variable, coordinate label, subsystem name, gauge slice, or record token.*

Proof. Let (X, s) be any quotient-respecting and separating clock presentation. Quotient respect makes s constant on $\sim_D^{\mathcal{U}}$ -classes, so s factors through $q_D^{\mathcal{U}}$. Separation makes the induced map injective, and surjectivity onto the presented state space is part of presentation. The induced bijection is unique because $q_D^{\mathcal{U}}$ is surjective. This is the clock specialization of standing-quotient uniqueness and finality [Maley, 2026c]. $\square$

5.5 Clock-role assignment

Let $L_{\text{Clk}}^{\mathcal{U}}$ denote the role “the physical reference whose readouts condition or order the target content for use $\mathcal{U}$.” A role assignment is a partial map

$$R_{\text{Clk}}^{\mathcal{U}} : L_{\text{Clk}}^{\mathcal{U}} \rightharpoonup \mathcal{Q}_{\text{Clk}}(D, \mathcal{U}). \quad (20)$$

It fixes a physical clock class, not a skin representative.

5.6 Operational example: a finite-resolution atomic clock

Consider an atomic transition counter used to timestamp detector records. The atomic species, transition, coupling to the readout apparatus, calibration convention, and stable-operation domain form the anchor signature. Retained tick records and their correlations with detector events form invariant clock-use profile content. Choice of zero, unit notation, and equivalent coordinate labeling are skin. Finite resolution, drift, and backreaction belong to Perf when they affect timestamping. If the transition becomes unreadable or the calibration ledger is lost, the candidate exits the certified class for that use even though the atom and Hamiltonian remain.

6 Clock-standing certificate criterion and canonical validity conditions

Definition 6.1 (Physical clock adequacy package). A candidate Θ satisfies the adequacy package for $\text{ClockStand}_D^U(\Theta)$ when:

- (C1) **Internality or declaration.** The candidate is internal to D or explicitly declared as background/effective anchor.
- (C2) **Constraint compatibility.** Gauge or redescription motion is not reclassified as physical clock evolution.
- (C3) **Physical descent.** Relational and readout constructions define elements of Q_D .
- (C4) **Use-adequate readout.** A readout relation exists for U ; a purely formal variable need not satisfy this clause for other purposes.
- (C5) **Reusable record content.** When U requires records, readouts can be retained as the same content under lawful redescription.
- (C6) **Coherent continuation.** Retained clock histories remain eligible at every stage; failed prefixes are not repaired as the same lineage.
- (C7) **No undeclared selector.** No convenience ranking, typicality rule, probability, representative, or foliation preference fixes the role without declared physical load.
- (C8) **Declared validity and failure domain.** Locality, monotonicity, periodicity, approximation order, backreaction, and breakdown conditions are stated.

6.1 Canonical local-clock conditions

Where a classical clock is used to gauge-fix first-class constraints C_I , the following local conditions are natural audit witnesses:

- (K1) the clock functions T^A have a nondegenerate bracket matrix $M^A_I = \{T^A, C_I\}$ on the patch;
- (K2) the surfaces $T^A = \tau^A$ intersect the relevant gauge orbits locally;
- (K3) the intersections are unique on the declared patch, or residual copies are explicitly quotiented;
- (K4) the corresponding complete observables are regular on that patch;
- (K5) turning points, zero determinant, and Gribov-type failures define patch boundaries.

For a single constraint, the familiar local condition is $\{T, C\} \neq 0$. These conditions are not universal definitions of time; they are concrete canonical witnesses of local clock eligibility.

6.2 Quantum clock conditions

A quantum implementation must additionally state, where relevant:

- (Q1) the physical state space or rigging/group-averaging map;
- (Q2) the physical inner product used to normalize conditional states;
- (Q3) the operator domain, self-adjointness, or POVM status of the clock readout;
- (Q4) covariance of the readout under the physical constraint;
- (Q5) periodicity, degeneracy, finite-dimensionality, nonideality, and backreaction limits;
- (Q6) whether different clock choices are related by a unitary/isometric physical map or define different theorem objects.

These items prevent a kinematic conditioning formula from being overreported as a physical clock theorem.

6.3 Monotonicity and record order are insufficient

Monotonicity can permit local inversion along a representative orbit, but does not by itself establish gauge invariance, a physical readout, or record reuse. Likewise, an ordered trace need not identify the same continuation locus or clock role. A formal clock can be monotonic and still fail the physical clock-standing use; a physical record clock can be locally valid without a global monotonic scalar.

Lemma 6.2 (Monotonicity insufficiency). *If T is monotonic on a representative history but its conditioned constructions do not define elements of $\mathbf{Q}_D$ or a readout adequate for $\mathbf{U}$, monotonicity alone does not imply $\text{ClockStand}_D^{\mathbf{U}}(T)$.*

Proposition 6.3 (Use-relative bivalence with graded quality). *For fixed D and $\mathbf{U}$, clock eligibility is bivalent on the evaluated fragment. Accuracy, resolution, stability, and backreaction may be graded inside the admitted model.*

Proof. A quality measure either affects eligibility for $\mathbf{U}$ or it does not. If it does, the declared threshold or domain induces a bivalent eligibility predicate; if it does not, it remains interior tensor data or bookkeeping. No intermediate boundary standing follows. $\square$

Theorem 6.4 (Clock-Standing Certificate Criterion). *Any admissible realization of a physical clock for use $\mathbf{U}$ on D supplies the adequacy package, native physical-descent certificate, use-adequate readout or record ledger, coherent continuation, and declared failure boundary. Failure of any required coordinate means that the submitted candidate does not instantiate the specified physical clock-use target on that domain.*

Proof. The target treats a readout or conditioned observable as determinate reusable physical content. Constraint compatibility and descent fix the physical object; the readout and record clauses fix the use; coherent continuation preserves the retained lineage; the no-selector clause blocks representative privilege; and the failure boundary fixes where the judgment is defined. Removing a required coordinate removes the corresponding clock-use role rather than producing a weaker realization of the same target. $\square$

7 Clock-standing certificate and audit algorithm

The formal definitions acquire model-facing content through the following reporting standard. The certificate is the positive construction required before clock-choice closure can be claimed.

Clock-Standing Certificate

A candidate Θ submitted to use U should report:

1. **Use:** the task the clock is meant to perform.
2. **Domain:** the sector, patch, approximation, or boundary conditions in force.
3. **Candidate and readout:** the formal variable/subsystem and its physical reading map.
4. **Constraint relation:** brackets, gauge fixing, covariance, or physical-state condition.
5. **Physical descent:** the quotient, operator, rigging, or history object carrying the construction.
6. **Record ledger:** how readings are fixed, retained, and reused when the use requires records.
7. **Continuation:** what counts as coherent clock evolution or conditioning.
8. **Failure boundary:** turning points, degeneracy, nonideality, approximation error, backreaction, or boundary breakdown.
9. **Rival-clock and selection status:** theorem-object, eligibility, anchor, quotient-profile divergence, physical equivalence, and any separate selection-governance classification.
10. **Terminal status and class:** richer scope, fail-closed, anchor, tensor, or skin, together with the use-relative class $[\Theta]_D^U$ when certified.

Clock Audit Algorithm

Input: fixed domain D , declared use U , and candidate Θ .

Step 1. Test theorem-object preservation. If the constraint object, quotient, physical state space, boundary data, or use changes, return RichScope.

Step 2. Test use eligibility. If constraint compatibility, physical descent, readout/record adequacy, or failure typing is missing, return Fail.

Step 3. Test definedness load. If varying Θ changes whether the use is well defined, return Anchor.

Step 4. Test physical-profile load. If definedness is fixed but relational observables, records, continuation, predictions, or failure behavior change, return Tensor.

Step 5. Otherwise return Skin.

Output: one terminal role/disposition, the certified clock class when applicable, and the data needed for rival-clock closure.

The algorithm is an ordered normal-form decision protocol for a declared comparison class. It is neither a numerical fitting procedure nor a heuristic priority rule, and it does not select the

highest-quality clock. Probabilities, Born weights, accuracy scores, and typicality measures may be legitimate physical content inside an admitted model; they are excluded only when used as undeclared authority to decide which physically equivalent clock has standing.

8 Physical descent, record fixation, and conditioned verdicts

8.1 Physical quotient descent

Definition 8.1 (Physical descent). A relational observable, conditional state, or clock readout descends when it is invariant under the declared redundancy relation and defines an element of $\mathbf{Q}_D$. On a regular smooth quotient this means

$$\mathcal{O}_{\text{Rel}}^\Theta = \tilde{\mathcal{O}}_{\text{Rel}}^\Theta \circ \pi, \quad \mathcal{R}_{\text{Clk}}^\Theta = \tilde{\mathcal{R}}_{\text{Clk}}^\Theta \circ \pi. \quad (21)$$

In a quantum implementation the same role may be carried by a physical operator, rigged observable, covariant POVM, or physical history class.

Lemma 8.2 (Representative-dependence diagnosis). *If two representatives of one physical object yield different clock-conditioned verdicts while all declared anchors and physical data are fixed, then either the construction fails descent or the theorem object omitted standing-bearing structure.*

8.2 Base clock-record profile

To avoid defining a quotient partly by the downstream verdict later tested for factorization, the base record profile excludes that verdict.

Definition 8.3 (Base clock-record profile). For an admissible clock-record assignment ρ , define

$$\text{Prof}_{\text{ClkRec}}^0(\rho) = (\text{Prof}_{\text{MR}}(\rho), \text{Ord}_\rho, \text{Cont}_\rho, \text{Fail}_\rho), \quad (22)$$

where Prof_{MR} is the imported measurement-record profile, Ord_ρ is record order, Cont_ρ is retained lineage, and Fail_ρ is failure typing.

Definition 8.4 (Base clock-record equivalence).

$$\rho \approx_{\text{ClkRec}}^0 \sigma \iff \text{Prof}_{\text{ClkRec}}^0(\rho) = \text{Prof}_{\text{ClkRec}}^0(\sigma). \quad (23)$$

The quotient $\mathcal{A}_{\text{ClkRec}} / \approx_{\text{ClkRec}}^0$ is the coarsest quotient preserving the declared base profile.

Theorem 8.5 (Conditioned-verdict factorization criterion). *Let $V : \mathcal{A}_{\text{ClkRec}} \rightarrow \mathcal{V}$ be a downstream conditioned verdict not included in the base profile. The verdict is attributable to the fixed theory-level clock construction iff there is a unique map*

$$\tilde{V} : \mathcal{A}_{\text{ClkRec}} / \approx_{\text{ClkRec}}^0 \longrightarrow \mathcal{V} \quad (24)$$

such that $V = \tilde{V} \circ q$, where q is the quotient map.

Proof. If V is constant on every base-profile class, the quotient universal property gives the unique factorization. If not, two assignments indistinguishable by the fixed record, order, continuation, and failure data yield different downstream verdicts. The verdict is then assignment-relative or depends on omitted physical structure; it cannot be attributed to the fixed theory-level clock construction alone. $\square$

8.3 Boundary-trace fixation as a conditional certificate

Definition 8.6 (Clock boundary trace). A clock boundary trace is a declared invariant predicate

$$\text{Tr}_{\text{Clk}} : \mathcal{T}_D / \equiv_{\text{Clk}}^{\cup} \longrightarrow \{0, 1\} \quad (25)$$

whose physical construction, domain membership, stability, and outcome independence are supplied as premises.

Boundary-trace admissibility certificate. A boundary trace fixes a clock class only when all six conditions hold:

- (T1) the trace is declared before the selection act;
- (T2) it is independently admitted rather than introduced in order to choose the clock;
- (T3) it is stable under lawful redescription;
- (T4) it is outcome-independent relative to the candidate comparison;
- (T5) its trace-positive set is singleton over canonical clock-profile classes; and
- (T6) it does not choose a representative inside an already fixed equivalence class.

Failure of any condition places the route in scope enrichment, representative privilege, unsupported class privilege, or hidden-selector import rather than lawful same-scope fixation.

Proposition 8.7 (Boundary-trace clock-role sufficiency). *If the trace-positive set is exactly one clock-profile class, that class is fixed for the declared role. If it is empty or non-singleton, the trace supplies no clock-role selection.*

Proof. The singleton conclusion follows directly from the supplied predicate. The proposition does not derive the trace. Its physical origin and admissibility remain separate burdens. An empty or non-singleton positive set requires either no assignment or additional declared structure; an undeclared tie-breaker would be representative privilege or unsupported class privilege. $\square$

9 Metric time, physical time, and temporal orientation

9.1 Distinct theorem objects

The clock problem must be separated from temporal direction. We distinguish: (i) physical time as ordering of retained consequence inside a realized physical interior; (ii) clock standing as use-relative eligibility of a reference carrier; and (iii) standing monotonicity as the irreversible order on standing classes. A cyclic clock may have standing; a standing arrow may exist without one global metric clock; and proper time can be a valid local descriptor without being the primitive source of temporal direction.

Table 7: Synchronization of the two time non-primitivity papers.

Object	Primary paper	Role
AASC kernel K_D	Upstream kernel theorem	Necessity-instantiated structure in non-degenerate regimes: admissibility, standing, reference, and irreversibility.
Standing arrow $<_S$	Paper I	Strict non-invertible quotient direction induced by irreversible standing fixation.
Physical time role T_D	Paper I / realized-interior import	Ordered retained consequence inside the realized physical interior.
Effective arrows	Paper I	Record, thermodynamic, causal, quantum-measurement, psychological, and cosmological projections or refinements.
Physical clock quotient $\mathcal{Q}_{\text{Clk}}(D, \mathcal{U})$	Paper II	Certified use-relative quotient of clock-use behavior for a declared domain and use.
Clock-selection governance	Paper II	Classification of attempts to privilege one clock representative or one class in a plural quotient.

The table is a dependency map, not a reduction. A clock may represent or operationalize an admitted time role, and effective arrows may constrain orientation-sensitive clock profiles, but no object silently performs another object’s work.

9.2 Metric licensing and no backflow

Proper time, duration, causal order, and synchronization are metric-interior descriptors. They require standing-positive paths, events, calibration, and a licensed metric projection [Maley, 2026m,h]. A successful proper-time or semiclassical construction cannot retro-authorize the constraint domain that licensed it. Empirical accuracy can support interior model use, but is not primitive boundary standing.

Theorem 9.1 (No clock backflow). *No metric or relational clock success generates standing backward across the boundary of its own declared domain.*

Proof. Using later success to confer prior eligibility is either carrier laundering, scope transport, hidden selector work, or same-act repair. Each changes the target or violates the imported no-backflow and irreversibility conditions. $\square$

Theorem 9.2 (No clock-to-kernel or clock-to-arrow backflow). *For a fixed kernel-faithful clock-use package $(D, \mathcal{U})$, certification of a physical clock class $[\Theta]_D^{\mathcal{U}} \in \mathcal{Q}_{\text{Clk}}(D, \mathcal{U})$ does not generate the AASC kernel, standing monotonicity, the physical time role T_D , record asymmetry, thermodynamic direction, causal direction, quantum-measurement direction, or cosmological orientation. It certifies only a clock-use carrier inside the declared scope.*

Proof. A certified clock already presupposes a fixed domain and use, physical descent, identifiable readouts or records, coherent continuation, and a typed failure boundary. If certification generated admissibility, standing, reference, or irreversibility, a downstream clock construction would become a lower generator of the kernel, contradicting the upstream necessity theorem. Using certification

to authorize standing monotonicity would reverse the dependency order. Using it to authorize T_D would promote one readout carrier into the role architecture that licenses retained-consequence ordering. Using it to authorize an effective arrow would either privilege a representative, install an undeclared selector, or change scope. Therefore a physical clock may represent, sample, or condition already licensed temporal roles, but it cannot ground the kernel or any temporal arrow. $\square$

9.3 Orientation-sensitive profiles

Proposition 9.3 (Clock reparametrization criterion). *A reparametrization $\tau \mapsto f(\tau)$ is skin only when it preserves every orientation-sensitive part of the full physical profile, including record order, positive-frequency or superselection sector, boundary conditions, causal orientation, and conditioned observables. Otherwise it is anchor or tensor change.*

This qualification is essential. Formal reversal of a label is not automatically physical time reversal, but neither is it automatically skin when the theory distinguishes frequency, boundary, causal, or thermodynamic orientation.

9.4 Three synchronization examples

Example 9.4 (Cyclic clock). A cyclic clock can have standing for a local phase-ordering or synchronization use even though it supplies no global linear parameter. Its local certification belongs to $\mathcal{Q}_{\text{CLK}}(D, \mathcal{U})$; the standing arrow remains the invariant non-invertible orientation of fixed consequence and is not generated by the cycle.

Example 9.5 (Noisy atomic clock). A finite-resolution atomic clock may drift, decohere, or lose precision. Those variations enter the performance profile or cross a declared failure boundary. They do not alter the standing arrow and do not create primitive temporal authority.

Example 9.6 (Semiclassical WKB time). A WKB parameter can be a lawful effective clock inside a declared heavy/light split, branch, gauge, inner-product, approximation, and error package. Its success does not retro-authorize an external time for the exact constrained theory; it realizes a scoped clock class.

Part III

Clock-Standing Theorems

10 No same-scope external-clock import

Definition 10.1 (Undeclared external clock). An undeclared external clock is a parameter t_{ext} used to generate or order physical continuation while being absent from the declared physical quotient object, boundary Hamiltonian data, background anchors, and readout ledger.

Theorem 10.2 (No same-scope external-clock import). *Let D be a fixed constrained clock domain and $\mathcal{U}$ a physical clock use. A temporal parameter external to the fixed theorem object cannot be silently imported as a same-scope clock for $\mathcal{U}$. It can carry clock standing only by being represented in the physical theorem object or explicitly declared as background, boundary, asymptotic, or effective anchor; that declaration changes or enriches the theorem object's anchor ledger.*

Proof. If the parameter changes no physical observable, readout, boundary charge, or continuation, it is presentation skin. If it changes physical continuation while remaining outside $\mathbf{Q}_D$ and the declared anchor ledger, the physical state description is incomplete or the parameter supplies hidden selector/authorizer load. If it is added explicitly as boundary Hamiltonian time, asymptotic time, deparametrizing matter, or effective background, the theorem object or anchor ledger has been enlarged. These cases exhaust the status of an external parameter on the fixed target. $\square$

Remark 10.3 (Boundary and asymptotic time). The theorem does not deny physical evolution generated by boundary Hamiltonians or asymptotic charges. Such structures are standing-bearing precisely because they are declared in the theorem object. The claim concerns an external parameter that carries physical force while remaining absent from that object.

Corollary 10.4 (Ordinary Schrödinger time). *Ordinary Schrödinger time may be a valid background anchor in a non-gravitational or semiclassical regime. It is not automatically transported into the fully constrained gravitational scope without a declared embedding or reduction certificate.*

11 Relational observables and physical quotient descent

Relational approaches seek physical predictions in correlations rather than evolution relative to an external parameter [Rovelli, 1990, 2002, Dittrich, 2006, Thiemann, 2006]. The standing burden is common even when the native mathematical implementation differs.

Definition 11.1 (Representative relational construction). A representative construction F_τ^Θ is a kinematic, gauge-fixed, conditional, or operator-valued object intended to express the value of F relative to clock reading τ .

Theorem 11.2 (Relational-observable descent theorem). *A representative construction is eligible as physical clock content only if it defines an element of the declared physical quotient object $\mathbf{Q}_D$. In a regular smooth reduction this means*

$$F_\tau^\Theta = \tilde{F}_\tau^{[\Theta]} \circ \pi. \quad (26)$$

In other implementations it means physical constraint invariance together with the relevant physical inner-product, operator-domain, group-averaging, or history-class conditions.

Proof. Physical content must be reusable as the same content under the declared redundancy relation. A representative-sensitive verdict is either nonphysical skin dependence or evidence that the theorem object omitted standing-bearing structure. Therefore the construction must descend. The theorem states a necessary condition; it does not guarantee that a global observable exists or that a formal expression is self-adjoint, normalized, or experimentally useful. $\square$

Corollary 11.3 (Gauge fixing is not sufficient). *A gauge-fixed expression can represent a physical relational observable only after descent, domain, and residual-copy conditions are established. Gauge fixing by itself does not confer physical clock standing.*

12 Clock-standing certificate necessity and presentation sufficiency

Definition 12.1 (Admissible physical clock presentation). An admissible presentation for use $\mathbf{U}$ supplies a declared clock role, an element of the physical quotient object, an adequate readout/record map for $\mathbf{U}$, coherent continuation, no undeclared selector, and an explicit validity/failure domain.

The following result is a target-adequacy theorem, not a dynamical existence theorem. It identifies conditions without which the specified physical clock-use role is not instantiated.

Theorem 12.2 (Clock-standing certificate necessity theorem). *Every admissible realization of $\text{ClockStand}_D^U(\Theta)$ satisfies*

$$\text{QuotDesc}_D(\Theta), \quad \text{RecReuse}_D^U(\Theta), \quad \text{CohCont}_D^U(\Theta), \quad \neg \text{HiddenSel}_D^U(\Theta), \quad (27)$$

together with a declared use domain and fail-closed boundary.

Proof. The physical use treats readouts or conditioned observables as determinate reusable contents. The imported kernel therefore requires target identity, eligibility, and no same-act repair. Physical identity under the constraint/redundancy structure requires descent to $\mathbf{Q}_D$. The use burden requires a readout or record relation adequate for $\mathbf{U}$. Retained use invokes trajectory coherence. A Metric no-selector discipline excludes an undeclared representative, probability, typicality, or convenience rule from fixing the role. Finally, a use-relative eligibility judgment is not well typed without its domain and failure boundary. $\square$

Proposition 12.3 (Presentation sufficiency). *A candidate supplied with the structures in equation (27), a declared role, and a declared validity domain constitutes a well-typed physical clock presentation by the definitions of this paper.*

Proof. The proposition is a typing result, not a dynamical existence theorem. The supplied data instantiate the declared clock-use target and permit the candidate to be evaluated by the ordered normal-form classifier. No claim follows that such data can always be constructed, that the clock is global, or that it is empirically optimal. $\square$

Remark 12.4 (No definitional inflation). The theorem is the necessity direction. The converse is intentionally not reported as an independent physical theorem, since it follows from what counts here as a complete physical clock presentation.

13 Frozen formalism as absence of an independently standing external generator

13.1 Qualified target

The stationary constraint equation does not have one universal interpretation. The following classification is limited to a domain in which the Hamiltonian constraint is treated as first-class redundancy and no boundary Hamiltonian, external temporal anchor, or deparametrizing clock has been declared.

Theorem 13.1 (Frozen-formalism reclassification). *On the qualified target above, a constraint equation such as $\widehat{\mathcal{H}}\Psi = 0$ does not supply external-parameter evolution on the physical state object. It expresses the absence, on that domain, of an independently standing external evolution parameter. It does not imply the absence of relational change, records, conditional observables, or effective time.*

Proof. The state is selected by the constraint rather than evolved by an external generator represented on the same physical object. Introducing an external parameter without adding it to the physical or boundary data would be hidden load. Relational observables, conditional states, histories, and effective clocks can still encode physical change when they satisfy descent and use-adequacy conditions. Thus external evolution is absent from the declared object, not all change from the theory. $\square$

Remark 13.2 (Deparametrized and boundary-generated evolution). If a constraint can be solved for a physical clock momentum, or if a nonvanishing boundary Hamiltonian generates physical asymptotic evolution, the corresponding time structure is part of a different or enriched theorem object. The reclassification does not deny that evolution.

14 Clock-pair physical normal form

The physical relation between two clock candidates is classified before any attempt to privilege one candidate is considered. Selection is a separate governance act treated in [section 15](#). “Submitted to one package” means submitted under a common declared use and comparison schema; the first branch tests whether a candidate’s standing conditions in fact preserve that theorem object. A candidate requiring a different quotient, physical Hilbert space, inner product, boundary charge, observable algebra, continuation law, or boundary signature is theorem-object divergent rather than same-package variation.

Definition 14.1 (Clock-pair comparison loci). For candidates submitted to one package $\mathfrak{C}_D^U$, compare in order:

- (L1) **Theorem-object locus:** constraint/solution object, redundancy relation, physical quotient, observable algebra, physical Hilbert or inner-product data where applicable, boundary charges, and continuation law.
- (L2) **Eligibility locus:** membership in Cert_D^U .
- (L3) **Anchor locus:** equality of A_D^U .
- (L4) **Canonical clock quotient:** equality of $q_D^U(\Theta)$ and $q_D^U(\Phi)$, equivalently equality of the invariant clock-use profile on fixed anchors.

Lemma 14.2 (Clock-pair divergence-locus lemma). *Every physically relevant relation between candidates submitted to one clock-use target changes at least one of the four loci above. If none changes, the candidates are the same physical clock for U up to skin.*

Proof. The four loci fix the object of comparison, admission to the use, definedness of the use, and all admitted physical clock-use consequences. By the Clock-Profile Finality Theorem, no same-scope physical functional distinguishes candidates agreeing on all four. A purported distinction changing none is therefore skin or an inadmissible representative-only discriminator. $\square$

Definition 14.3 (Clock-pair decision order). For $\Theta, \Phi \in \mathcal{C}_D^U$ evaluate:

- (D1) If the theorem object differs: theorem-object divergence.
- (D2) Otherwise, if one candidate is not certified: eligibility divergence or clock-use failure.
- (D3) Otherwise, if the anchor signatures differ: anchor divergence.
- (D4) Otherwise, if the canonical quotient classes differ: quotient-profile/tensor divergence.
- (D5) Otherwise: physical clock equivalence for U .

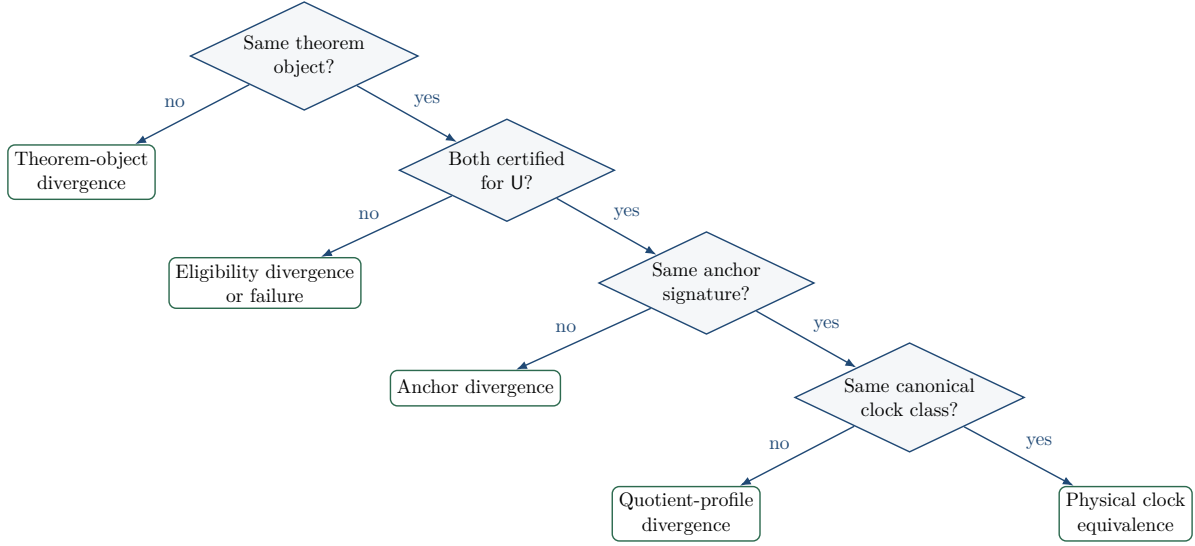


Figure 2: Physical normal form for a clock-candidate pair. Selection authority is audited separately.

Theorem 14.4 (Clock-Pair Physical Normal-Form Theorem). *Under the declared decision order, every clock-candidate pair receives exactly one physical classification: theorem-object divergence, eligibility divergence/failure, anchor divergence, quotient-profile/tensor divergence, or physical clock equivalence. No sixth same-scope physical pair relation survives.*

Proof. The decision order is single-valued, and coverage follows from [theorem 14.2](#). Performance, stochasticity, finite resolution, drift, decoherence, backreaction, and approximation error are not additional pair relations. If they change certification they enter eligibility/failure; if they change definedness they enter anchor divergence; if they change admitted physical behavior they enter quotient-profile divergence; if they change none they are skin. $\square$

Corollary 14.5 (Clock-dependent predictions). *Clock-dependent predictions are not an untyped contradiction. They arise from theorem-object divergence, eligibility divergence, anchor divergence, or quotient-profile divergence unless the candidates are physically equivalent. Any attempt to privilege one candidate is a separate selection-governance question.*

15 Clock-selection governance and choice typing closure

Definition 15.1 (Clock-selection act). A clock-selection act is a partial map

$$S : (D, U, K) \multimap \text{Cert}_D^U, \quad (28)$$

where $K \subseteq \text{Cert}_D^U$ is a candidate family and S is claimed to fix the occupant of the clock role for U .

Definition 15.2 (Hidden selection). “Hidden selection” is an umbrella term for representative privilege, unsupported class privilege, or impossible authority acquisition when the selection act is not declared as a theorem-object change.

Success, simplicity, convenience, accuracy, stability, empirical adequacy, probability, typicality, and utility may be legitimate internal performance data, experimental-design criteria, model-use reasons, or approximation controls. They become governance-relevant only when promoted to same-scope standing-generating authority or singular class selection, in which case declared anchor, quotient-profile, or prior boundary-trace support is required.

Theorem 15.3 (Clock-Selection Governance Theorem). *Under the declared decision order, every attempted singular clock selection receives the first applicable terminal governance status:*

- (S1) **Declared role fixation.** *Non-trace anchor or quotient-profile/tensor content already present in the clock package fixes one canonical clock class.*
- (S2) **Declared boundary-trace fixation.** *No non-trace role content fixes a class, but a stable, outcome-independent singleton boundary trace already declared in the package or independently admitted before the selection act fixes one class without selecting a representative.*
- (S3) **Scope enrichment.** *Selection requires a new clock field, external time, boundary trace, preferred foliation, state variable, criterion, boundary channel, or theorem-object component.*
- (S4) **Representative privilege.** *Selection chooses one representative inside a canonical clock class without admitted support.*
- (S5) **Unsupported class privilege / hidden selector.** *Selection chooses one class among plural canonical clock classes without declared anchor, quotient-profile, boundary-trace, or theorem-object support.*
- (S6) **Impossible authority acquisition.** *Selection is justified by success, simplicity, convenience, accuracy, utility, probability, typicality, explanation, consensus, repetition, empirical adequacy, cross-domain inheritance, repair, or another non-gating consideration acting as standing-generating authority rather than as declared anchor/quotient-profile content or admitted performance data.*

The first two are lawful role-fixation routes inside the fixed package. The third is a declared new theorem object. The last three supply no admissible same-scope clock authority. Accuracy, stability, simplicity, and empirical success remain legitimate considerations for instrument design, approximation choice, model use, and performance comparison after admission; the theorem excludes only their promotion into undeclared authority for clock standing or singular class selection.

Proof. The order makes the statuses disjoint. If existing non-trace anchor or quotient-profile content fixes one class, (S1) applies. Only if it does not, an already-declared or independently pre-admitted singleton trace can fix the role by boundary-trace fixation, giving (S2) [Maley, 2026e]. A trace or selector introduced by the selection act changes the comparison class and gives (S3). If the quotient class is already fixed and one representative is chosen without support, the act is AMetric representative privilege, route R7, giving (S4). If the quotient is plural and one non-equivalent class is chosen without declared support, the act introduces a selector or second classifier, AMetric routes R1/R5, and gives (S5). Every remaining attempted source of authority is a generator, carrier, parameter, bridge, transfer, accumulation, or repair route and collapses under AMetric route exhaustion, the apex no-repair/no-generator/no-carrier theorems, and the Impossibility Suite [Maley, 2026b,r,k]. No seventh same-scope selection route remains. $\square$

Table 8: Clock-selection branches and their fixed-domain status.

Selection status	Closure source	Result
Declared role fixation	Existing anchor/tensor content	Lawful same-scope fixation
Declared boundary trace	Prior singleton trace	Lawful same-scope fixation
Scope enrichment	Comparison-class change / scope transport	New theorem object
Representative privilege	AMetric R7	Inadmissible same-scope selection
Unsupported class privilege	AMetric R1/R5; no competing gates/interiors	Hidden selector / second classifier
Impossible authority acquisition	AMetric R2–R4, R6, R8; no generator/carrier/repair/bridge	Inadmissible authority claim

Definition 15.4 (Clock-choice typing closure). The package has clock-choice typing closure when every candidate pair receives a physical normal-form classification and every attempted singular selection receives a governance classification.

Theorem 15.5 (Physical-Clock Quotient Finality and Choice-Closure Theorem, kernel-faithful exhaustion form). *For a fixed package $\mathfrak{C}_D^U$ satisfying the imported AASC/AER conditions:*

- (C1) *the declared package instantiates the non-degenerate target and remains kernel-faithful;*
- (C2) *$\mathcal{Q}_{\text{CLK}}(D, U)$ is final under every admissible same-scope clock-use functional;*
- (C3) *every clock-candidate pair has the physical normal form of [theorem 14.4](#);*
- (C4) *every singular clock-selection act has the ordered governance normal form of [theorem 15.3](#);*
- (C5) *no residual same-scope clock identity, physical relation, or selection authority exists outside the canonical quotient and declared anchor structure; and*
- (C6) *the closure is clock-finality rather than kernel-generation or arrow-generation: it neither rederives nor replaces the AASC kernel, standing monotonicity, physical time, or the effective-arrow taxonomy of Paper I.*

Equivalently, there is no admissible residual relation R_{res} or selector S_{res} that fails to factor through those structures, and no clock theorem is promoted into a primitive source of temporal direction.

Proof. Completeness follows from the fixed clock-use comparison class. Any candidate difference first either preserves or alters the theorem object. If it alters the constraint object, physical quotient, physical state space, inner product, boundary data, observable algebra, continuation law, or declared use, it is theorem-object divergence or scope enrichment. If it preserves the theorem object, it either affects certification, definedness and anchor conditions, quotient-profile behavior, selection authority, or none of these. Those possibilities are exactly eligibility divergence or failure, anchor divergence, quotient-profile divergence, the ordered selection-governance branches, or physical equivalence and skin. No sixth same-scope clock relation and no seventh same-scope selection route remain.

Clause (C1) fixes the target-instantiation and dependency posture. Clause (C2) is the Clock-Profile Finality Theorem, with future functionals restricted to lawful continuation inside the fixed probe package. Clause (C3) follows from finality and the pair divergence-locus lemma. Clause (C4)

follows from quotient finality, declared or independently admitted boundary-trace fixation, and the impossibility closure against generators, carriers, accumulation, repair, and bridge principles. Clause (C5) follows because any alleged residual relation would distinguish canonical-equivalent clocks or add an unrepresented physical functional, while any alleged residual selector would have to use declared role content, invoke a prior boundary trace, enrich scope, privilege a representative, privilege one plural clock class without support, or acquire authority through a forbidden route. Clause (C6) follows from [theorem 9.2](#): quotient finality closes clock identity and choice only, while the kernel theorem and predecessor arrow theorem retain authority over their upstream targets. Hence no residue survives. $\square$

15.1 Unique admissible interior and typed clock plurality

Proposition 15.6 (Interior uniqueness does not collapse clock plurality). *For a fixed standing-instantiated clock domain, the admissible interior and gate classification are unique, while the physical clock quotient may contain zero, one, or several canonical clock classes. A plural quotient records declared anchor or quotient-profile/tensor differences inside the unique admissible interior; it is not plurality of admissibility regimes.*

Proof. The AMetric apex closure and *Unus* eliminate disagreeing same-domain gates and discrete multiplicity of admissible interiors [[Maley, 2026b,r](#)]. The clock quotient is formed only after that interior is fixed. Distinct quotient classes therefore differ by admitted clock-use content within the one interior. Introducing a second gate or clock-governance regime to choose among those classes would violate gate uniqueness, interior uniqueness, and selection closure. $\square$

15.2 Closure, realization, singularity, and plurality

Definition 15.7 (Clock-choice closure). $\text{Closed}_{\text{Clk}}(D, U)$ means that the final quotient, pair normal form, and selection-governance theorem all hold on the full candidate class $\mathcal{C}_D^U$.

Definition 15.8 (Clock realization statuses).

$$\text{Realized}_{\text{Clk}}(D, U) \iff \mathcal{Q}_{\text{Clk}}(D, U) \neq \emptyset, \quad (29)$$

$$\text{Singular}_{\text{Clk}}(D, U) \iff |\mathcal{Q}_{\text{Clk}}(D, U)| = 1, \quad (30)$$

$$\text{Plural}_{\text{Clk}}(D, U) \iff |\mathcal{Q}_{\text{Clk}}(D, U)| > 1. \quad (31)$$

Corollary 15.9 (Clock-quotient status trichotomy). *For every closed set-sized clock-use package (or the corresponding class-theoretic analogue when the candidate collection is not set-sized), exactly one quotient status holds:*

$$\mathcal{Q}_{\text{Clk}}(D, U) = \begin{cases} \emptyset, & \text{no certified physical clock for } U, \\ \{[\Theta]\}, & \text{one physical clock class for } U, \\ \{[\Theta_i]\}_{i \in I}, |I| > 1, & \text{typed physical clock plurality.} \end{cases} \quad (32)$$

These are three possible structures of the clock quotient inside one fixed standing-governed admissible interior, not three interiors. A plural quotient contains genuine anchor or quotient-profile differences. Closure types that plurality; it does not create an undeclared preferred class or a second clock-governance regime. A plural quotient is therefore not residual ambiguity left by the theorem but one of its closed outputs. Any request for a further preferred class is a new selection act governed by [theorem 15.3](#).

15.3 Saturated application registries

Definition 15.10 (Rival-registry saturation). For an application registry $\mathcal{R}_D^U \subseteq \mathcal{C}_D^U$, $\text{Sat}_D^U(\mathcal{R}_D^U)$ holds when every same-scope certified candidate generated by the declared candidate schema is represented in $\mathcal{R}_D^U$, canonical-clock-equivalent to a member, or covered by a proved schema branch whose first standing-relevant relation is represented in the registry. A registry is not saturated when a candidate is omitted by post-hoc narrowing, expansion, retagging, or other domain motion; such a move changes the audit rather than completing it.

Definition 15.11 (Saturated registry completion). A registry is complete relative to a certified candidate Θ iff

$$\text{RegComplete}_D^U(\Theta, \mathcal{R}_D^U) \iff \text{Sat}_D^U(\mathcal{R}_D^U) \wedge \forall \Phi \in \mathcal{R}_D^U, \text{Term}_D^U(\Theta, \Phi). \quad (33)$$

Corollary 15.12 (Saturated Clock-Use Completion). *A finite or otherwise explicit application audit warrants a completion claim only when its rival registry carries a saturation certificate and every registry member receives a terminal physical or governance classification. Empty or selectively truncated registries do not establish completion.*

Remark 15.13 (Universal closure, registry completion, and model audit). Universal clock-choice closure quantifies over the full candidate class $\mathcal{C}_D^U$. Registry completion is a finite or explicit application result and requires saturation. A worked model may establish realization and classify representative rivals without proving registry saturation. These are distinct reportable statuses.

Full-strength endpoint

The multiple-choice problem is converted into the canonical quotient $\mathcal{Q}_{\text{Ck}}(D, U)$. Relative to the declared package, the quotient may be empty, singleton, or physically plural. What closure excludes is not plurality, but any untyped same-scope clock relation or selection authority outside the quotient and declared anchors.

Why nonselection is a theorem output

The absence of a preferred clock in a plural quotient is not an incompleteness in the closure theorem. It is one of the theorem's possible final structures. A further preference is a new selection act and must be fixed by declared role content, a prior singleton boundary trace, or an explicit enrichment of scope. Without such authority, no preferred class survives.

16 Clock disagreement is not one phenomenon

The closure theorem separates physical clock-pair relations from governance acts that attempt to privilege one clock class.

Table 9: Physical clock-pair relations and selection-governance acts.

Case	What differs	Fixed-domain diagnosis
t versus $f(t)$	Only a lawful reparametrization	Skin, provided the full orientation-sensitive profile is preserved
Two finite clocks of different precision	Accuracy, noise, or disturbance	Admitted quotient-profile/tensor performance difference
Scalar clock valid only on a monotonic sector	Domain of definedness	Anchor difference or local failure boundary
Scalar time and volume time give different relational observables	Physical quotient/readout profile	Quotient-profile/tensor divergence
Clock choice changes physical Hilbert space or inner product	The theorem object	Richer-scope/theorem-object divergence
A rule chooses one representative within an equivalent clock class	No declared content distinguishes the representative	Representative privilege
A rule chooses one class from a plural quotient by “simplicity” or “naturalness”	No declared anchor, quotient-profile, or boundary trace supports the choice	Unsupported class privilege / hidden selector
A candidate reaches a turning point or loses a readout	Eligibility for the use	Fail-closed

Proposition 16.1 (Performance/standing separation). *Once a clock presentation is admitted for use $\mathcal{U}$, accuracy, precision, resolution, noise, decoherence, drift, and backreaction may vary continuously or probabilistically as interior physical data. They affect clock standing only when they cross a declared eligibility or failure condition for $\mathcal{U}$.*

Proof. Clock standing answers whether the readout role is well typed and reusable for the declared task. Performance quantities compare admitted implementations. If a performance value changes no eligibility judgment, it is quotient-profile/tensor quality data; if a declared threshold is crossed, the induced eligibility predicate is bivalent for that use. Thus approximate and noisy clocks are compatible with use-relative standing. Performance variation is not a seventh normal-form branch: it is quotient-profile/tensor divergence when it changes the admitted physical clock profile for $\mathcal{U}$, and skin when it changes no such profile. $\square$

Interpretive payoff

AASC closes the question that “which clock is really time” leaves untyped. It identifies the canonical physical clock quotient, classifies every pair relation inside it, and separately governs

any act that attempts to privilege one class or representative.

17 Clock failure, irrecoverability, and compatible local atlases

Definition 17.1 (Clock trajectory). A clock trajectory for $\mathcal{U}$ is a retained finite sequence of states together with clock readouts and conditioned physical content. It is coherent when every retained stage remains eligible for the same use and physical profile.

Theorem 17.2 (No deferred clock coherence). *If a retained prefix fails clock eligibility for $\mathcal{U}$, no continuation that preserves that prefix restores coherence for the same clock lineage.*

Proof. The failed stage remains in every extension. Reclassifying the extension as coherent would erase the act-time failure or change identity. The former violates irreversibility; the latter is reset or a new clock patch. $\square$

17.1 Clock-atlas compatibility criterion

A global scalar time need not exist. Let

$$\mathfrak{A}_{\text{Clk}} = \{(U_\alpha, [\Theta_\alpha]_D^\mathcal{U})\}_{\alpha \in A} \quad (34)$$

be local physical clock presentations. On overlaps, suppose maps $\Xi_{\alpha\beta}$ preserve the physical observable/readout algebras, base record quotient, continuation, and failure typing; are inverse up to profile equivalence; and satisfy a cocycle condition on triple overlaps.

Proposition 17.3 (Clock-atlas compatibility criterion). *Given local clock objects and overlap maps satisfying the stated preservation and cocycle conditions, the family defines a compatible atlas-level clock presentation. Numerical overlap of clock ranges alone is insufficient.*

Remark 17.4 (No existence theorem). The proposition is a certificate criterion. It neither proves that such transition maps exist nor that the resulting atlas is unique. Establishing a genuine gluing theorem requires a native category of physical observable/record objects and a descent proof in the selected quantum-gravity formalism.

18 Semiclassical and effective time

Semiclassical time is a successful effective construction, not a defective substitute. In Born–Oppenheimer or WKB treatments, a gravitational phase and heavy/light split support approximate matter Schrödinger evolution with controlled corrections [Kiefer and Singh, 1991, Kiefer and Wichmann, 2018, Kiefer, 2012].

Definition 18.1 (Semiclassical clock package).

$$\Sigma_{\text{sc}} = (\mathcal{S}_{\text{heavy}}, \mathcal{S}_{\text{light}}, \chi, \psi, \epsilon, \mathcal{G}_{\text{BO}}), \quad (35)$$

where the split, WKB branch, conditioned state, approximation order, and decomposition/gauge freedom are explicit.

Theorem 18.2 (Semiclassical-Time Scope Theorem). *Semiclassical time is a lawful physical clock presentation on a declared approximation domain when the heavy/light split, WKB branch, gauge, physical inner product, approximation order, and readout use are included in the effective theorem object. It is not an undeclared same-scope external time for the underlying exact constrained theory.*

Table 10: Semiclassical time as a successful effective clock package.

Element	Clock-standing role
Heavy gravitational background	Effective anchor
Selected WKB branch and orientation	Sector/boundary anchor
Light quantum subsystem	Conditioned quotient profile (ATS tensor)
Emergent Schrödinger parameter	Licensed effective clock readout
Higher-order corrections and backreaction	Quotient-profile/tensor performance and error ledger
Branch crossing or approximation breakdown	Declared failure boundary or scope transition

Proof. The effective parameter depends on the split, branch, phase, gauge, and approximation order. When declared, these are the physical data of the effective theorem object. When omitted while affecting evolution, unitarity, or corrections, they are hidden load. Breakdown of the approximation is a failure boundary, not evidence that effective time was illegitimate within scope. $\square$

A WKB time package realizes a clock class only relative to the selected branch, heavy/light split, background validity domain, physical inner product, error ledger, and failure boundary. Two WKB branches or decompositions with the same full profile are equivalent for the use; branches changing orientation, conditioned dynamics, inner product, or breakdown domain instantiate anchor, quotient-profile, or theorem-object divergence rather than an untyped ambiguity. Branch selection is lawful only when fixed by declared role content or a prior boundary trace; otherwise it is unsupported class privilege or scope enrichment.

19 Secondary result: ordered clock-candidate normal form

Lemma 19.1 (Clock-use divergence-locus lemma). *For an evaluated candidate submitted to a declared use U , the first standing-relevant difference from the fixed target occurs at one of four loci: theorem-object identity, use eligibility, use definedness, or the physical clock profile. If none changes, the candidate differs only by skin.*

Proof. A clock-use claim can change only by changing the object on which it is evaluated, whether the candidate is eligible, the preconditions under which the use is defined, or the quotient-level relational/readout/continuation/failure content delivered by the use. A variation changing none of these changes no physical clock-use fact. $\square$

Definition 19.2 (Ordered classifier). For a candidate Θ submitted to $\mathcal{U}$, define

$$\text{NF}_D^{\mathcal{U}}(\Theta) = \begin{cases} \text{RichScope}, & \Theta \text{ changes the theorem object;} \\ \text{Fail}, & \Theta \text{ remains same-scope but fails use eligibility;} \\ \text{Anchor}, & \Theta \text{ is admitted and varying it changes use definedness;} \\ \text{Tensor}, & \text{definedness is fixed but the physical profile changes;} \\ \text{Skin}, & \text{neither definedness nor the physical profile changes.} \end{cases} \quad (36)$$

The cases are evaluated in the displayed order.

Theorem 19.3 (Ordered Clock-Candidate Normal-Form Theorem). *Every evaluated candidate on the declared comparison class receives exactly one terminal output of $\text{NF}_D^{\mathcal{U}}$ under the displayed decision order. Within the same domain, admitted clock-related content is therefore anchor, quotient-level invariant content (the ATS tensor branch), or skin; richer scope and fail-closed are prior terminal dispositions. No additional same-scope clock-bearing role survives.*

Proof. The first test partitions theorem-object preservation. On the preserved object, the second partitions use eligibility. For eligible candidates, varying the candidate either changes use definedness or does not. With definedness fixed, it either changes the physical clock profile or does not. These binary decisions are exhaustive and mutually exclusive under the declared decision order. The ATS theorem identifies the three admitted branches as anchor, tensor, and skin. A purported further same-scope role would have to affect the use while changing neither definedness nor physical profile, contradicting the divergence-locus lemma. $\square$

Corollary 19.4 (No primitive preferred foliation). *A foliation is skin when physically inert, anchor/tensor when declared and physically relevant, richer scope when it changes the theorem object, or fail-closed when used as undeclared selector. It has no residual primitive role.*

Corollary 19.5 (No primitive best-clock functional). *A quality functional may compare already admitted clocks for a declared use. It does not generate eligibility or select among physically equivalent candidates at the boundary.*

Part IV

Applications and Standard Routes

20 Parametrized particle: parameter, gauge clock, and complete observable

The parametrized particle displays the difference between an arbitrary history parameter and a local internal clock. Let

$$S = \int d\lambda (p_q \dot{q} + p_t \dot{t} - NC), \quad \mathcal{C} = p_t + H(q, p_q) \approx 0. \quad (37)$$

The curve parameter λ and multiplier N are representative data. The canonical variable t is a formal relational clock candidate because

$$\{t, \mathcal{C}\} = 1, \quad (38)$$

so the gauge-fixing surfaces $t = \tau$ intersect each regular constraint orbit locally and uniquely in the simple model.

A complete observable can be written schematically as

$$Q_\tau = \alpha_s(q)|_{\alpha_s(t)=\tau}, \quad (39)$$

where α_s is the constraint flow.

Proposition 20.1 (Parametrized-particle audit). *The parameter λ is skin. The canonical variable t is a formal clock candidate, and the family Q_τ is eligible as physical clock content on patches where the gauge fixing is nondegenerate, unique, and quotient-descending. A physical readout use additionally requires a declared record or operational map.*

Proof. Changing λ changes only the orbit parametrization. The nonzero bracket makes $t = \tau$ a local gauge fixing; uniqueness on the patch prevents residual copies. The complete observable is constant on the constraint orbit and therefore descends to the reduced object. The physical readout clause is additional to the formal construction, in accordance with the formal/physical distinction of section 5. $\square$

Turning points, a vanishing gauge-fixing determinant, multi-valued intersections, or residual copies are clock-patch boundaries. They are not repaired by continuing the same failed patch.

21 Worked audit: scalar-field clock in minisuperspace

A two-dimensional minisuperspace model makes the certificate concrete. The section is a local clock-standing audit; complete quantization and global preferred-clock questions lie outside this target. Let $\alpha = \ln a$ denote a logarithmic scale variable and let ϕ be a homogeneous scalar field. After conventional rescalings, a broad schematic class has one Hamiltonian constraint

$$C = -p_\alpha^2 + p_\phi^2 + U(\alpha, \phi) \approx 0. \quad (40)$$

For the simplest massless-scalar sector with U independent of ϕ ,

$$\{\phi, C\} = 2p_\phi. \quad (41)$$

On a sector with $p_\phi \neq 0$ and fixed sign, the gauge condition

$$\chi_\tau := \phi - \tau = 0 \quad (42)$$

has local gauge-fixing determinant

$$\Delta_\phi := \{\chi_\tau, C\} = 2p_\phi \neq 0. \quad (43)$$

It therefore intersects the relevant constraint orbits transversely on the patch. This is a local nondegeneracy witness, not a global clock theorem. Physical clock standing further requires local uniqueness of the intersection and exclusion or explicit treatment of residual Gribov copies. A relational observable for a phase-space function F is written schematically as

$$O_{F,\phi}(\tau) = \alpha_s(F)|_{\alpha_s(\phi)=\tau}, \quad (44)$$

where α_s is the constraint flow.

21.1 Clock certificate for the classical patch

Table 11: Scalar-field clock certificate on a local minisuperspace sector.

Certificate coordinate	Model realization
Declared use	Construct $O_{F,\phi}(\tau)$ and order selected cosmological records
Domain	Connected sector with $p_\phi \neq 0$, fixed sign, and regular constraint flow
Clock candidate/readout	Scalar value $\phi = \tau$
Constraint relation	$\{\phi, C\} = 2p_\phi \neq 0$
Physical descent	Complete observables constant on the declared constraint orbits
Continuation	Evolution in the same monotonic sector
Failure boundary	$p_\phi = 0$, turning point, multiple orbit intersection, or loss of regularity
Terminal status	Quotient-profile/tensor local clock content on fixed patch anchors

Proposition 21.1 (Local scalar-clock standing). *On the declared regular sector, ϕ is a physical clock candidate for the relational-observable use when the gauge fixing is unique on the patch, the complete observables descend to the physical quotient, and any required readout or record map is supplied. The conclusion is local and use-relative; it does not identify ϕ with primitive or global time.*

Proof. Equation (41) supplies a local nondegeneracy witness, not a global clock theorem. The declared domain excludes points where the gauge fixing degenerates. Orbit invariance of the complete observable supplies classical descent. The fixed-sign sector, readout role, and failure conditions supply the anchor and continuation ledger. These are exactly the clock-profile necessity conditions for the declared use. $\square$

21.2 Quantum sector and positive-frequency anchor

For models admitting a Klein–Gordon-type separation with a massless scalar, including common Wheeler–DeWitt and loop-quantum-cosmology presentations, the physical constraint can be written in the schematic audit form

$$\partial_\phi^2 \Psi(\alpha, \phi) = -\hat{\Theta}_\alpha \Psi(\alpha, \phi), \quad (45)$$

with a positive operator $\hat{\Theta}_\alpha$ on the relevant domain. A positive-frequency sector then satisfies

$$-i\partial_\phi \Psi = \sqrt{\hat{\Theta}_\alpha} \Psi, \quad (46)$$

provided the self-adjointness, physical inner product, spectral domain, and sector choice are fixed [Ashtekar et al., 2006, Kiefer, 2012]. The equation exhibits a successful relational Schrödinger form; it also shows that the clock is not supplied by the symbol ϕ alone. If $\sqrt{\hat{\Theta}_\alpha}$ is self-adjoint on the declared physical domain, the positive-frequency evolution preserves an inner product of the form

$$\langle \Psi_1, \Psi_2 \rangle_{\text{phys}} = \langle \Psi_1(\cdot, \phi_0), \Psi_2(\cdot, \phi_0) \rangle_{\mathcal{H}_\alpha}, \quad (47)$$

independent of the reference value ϕ_0 within that sector. The physical Hilbert space, positive-frequency choice, self-adjoint extension, and spectral domain are therefore clock anchors or quotient-profile/tensor load, not consequences of the symbol ϕ alone.

Proposition 21.2 (Quantum scalar-clock audit). *The scalar variable has quantum clock standing only relative to the physical inner product, frequency/sector choice, self-adjoint operator domain, and conditioned observable family. Mixing of frequency sectors, a change of self-adjoint extension, or a rival physical inner product can produce anchor divergence, quotient-profile/tensor divergence, or theorem-object change rather than mere reparametrization.*

21.3 Rival-clock comparison

Compare ϕ with a volume or scale variable v submitted to the same relational-observable use. The decision order is concrete. First, if v -conditioning requires a different physical Hilbert space, inner product, self-adjoint extension, boundary condition, or constraint implementation, the pair is theorem-object divergent. Second, if the theorem object is fixed but v reaches a turning point, becomes noninvertible, or loses its readout while ϕ remains certified, the first divergence is eligibility/-failure. Third, if both remain certified but their monotonic sectors or validity patches differ, the first divergence is anchor divergence. Fourth, if anchors agree but their complete observables, conditioned states, record order, or singularity verdicts differ, the first divergence is quotient-profile/tensor divergence. Only when every admitted probe agrees in definedness and value do ϕ and v belong to the same physical clock class. Thus the normal form yields:

- identical complete-observable, readout, continuation, performance, and failure behavior: physical clock equivalence;
- different monotonic sectors or validity patches: anchor divergence;
- different relational observables or singularity verdicts on fixed anchors: quotient-profile/tensor divergence;
- different physical inner products, self-adjoint extensions, boundary data, or constraint implementations: theorem-object divergence;
- a turning point, noninvertible gauge fixing, or readout loss: eligibility failure.

A separate preference for the “simpler” variable is representative privilege when the candidates belong to one clock class, unsupported class privilege when they are distinct classes in a plural quotient, or impossible authority acquisition when simplicity or success is treated as an authorizer. No residual “better clock” category remains: practical superiority is admitted performance content, while singular authority remains a separate governance act. Clock-dependent singularity and unitarity results in quantum cosmology provide concrete instances of these alternatives [Gielen and Menéndez-Pidal, 2020]. The conclusion is not that ϕ is the real time. It is that ϕ -time has a declared profile, a validity patch, a failure boundary, and a classified relation to rival clocks.

21.4 Realization and closure status of the scalar-clock package

Proposition 21.3 (Local scalar-clock realization). *On a monotonic sector where $\{\phi, C\} \neq 0$, the gauge slice is locally nondegenerate, relational observables descend, and the readout/record and quantum-sector data are fixed, the scalar field determines a realized local clock class*

$$[\phi]_D^{\mathbf{U}_\phi} \in \mathcal{Q}_{\text{Clk}}(D, \mathbf{U}_\phi). \quad (48)$$

The bracket condition is a local nondegeneracy witness, not a global clock theorem; the declared domain excludes points where the gauge fixing degenerates. Outside that sector the candidate leaves the certified class rather than becoming a global time.

Proposition 21.4 (Scalar-clock closure status). *The scalar field realizes a local physical clock class on the declared monotonic sector. It does not by itself establish saturated completion of the model’s full clock problem. Every same-scope rival to the realized scalar-clock class is typed by the universal Clock-Quotient Finality and Choice-Closure Theorem; singular clock fixation requires $|\mathcal{Q}_{\text{Clk}}(D, \mathcal{U}_\phi)| = 1$ or an independently licensed boundary/anchor fixation theorem. A finite rival table warrants a scope-complete claim only when its registry is saturated in the sense of [theorem 15.12](#).*

Table 12: Closure classification for rivals to the scalar clock.

Rival-clock relation	Closure classification
Reparametrization of ϕ preserving the full profile	Same clock class $[\phi]_D^{\mathcal{U}_\phi}$
Scale or volume clock with a different validity sector	Anchor divergence
Rival clock yielding different relational observables or singularity verdicts	Quotient-profile/tensor divergence
Clock choice changing the physical inner product or self-adjoint extension	Theorem-object change
Preference for one representative within the scalar clock class	Representative privilege
Preference for one non-equivalent clock class by simplicity or naturalness	Unsupported class privilege / hidden selector
Empirical success or explanatory power treated as primitive clock authority	Impossible authority acquisition / carrier attempt
Loss of monotonicity, descent, or readout reuse	Fail-closed

The scalar-clock example therefore demonstrates realization without illicit uniqueness: certification identifies a local quotient class, the AMetric comparison-class lock determines whether a proposed rival is genuinely same-scope, the universal closure theorem types every such rival, and *Unus* blocks introduction of a second same-domain clock-governance regime to privilege one class. Singular fixation requires a singleton quotient or an independently licensed role-fixation certificate.

The example also displays the exhaustion-first strength of the method. AASC does not choose between ϕ , volume, proper time, or a WKB parameter by convenience. It fixes the theorem object, tests certification, compares anchors and quotient profiles, and then governs any singular preference. Once those loci are exhausted, no residual clock-choice category remains.

22 ADM gravity, lapse/shift, boundaries, and many-fingered evolution

Canonical GR has

$$H_{\text{tot}}[N, N^i] = H[N] + H_i[N^i] + H_{\partial\Sigma}, \quad (49)$$

where the boundary term is absent for suitable compact cases and otherwise encodes the boundary conditions and physical charges required for differentiability. The AER classification identifies the

bulk constraint surface and characteristic distribution with the constrained theorem object [Maley, 2026a,j]. The classification of coordinate time, lapse, and shift as skin or presentation is made only after boundary charges, asymptotic Hamiltonians, and external clock anchors have been fixed or excluded from that theorem object.

Proposition 22.1 (Lapse/shift classification with boundary guard). *Bulk lapse and shift variations that preserve all declared boundary data, asymptotic charges, and reduced observables are presentation variables on the fixed ADM/AER object. Boundary lapse/shift data associated with a physical Hamiltonian or asymptotic symmetry are anchor or tensor content and are not skin.*

Proof. On the fixed bulk constraint object, changes that move only along the characteristic and redescription distribution alter no reduced observable and are skin. When boundary values determine a nonzero Hamiltonian, energy, momentum, or asymptotic symmetry action, the variation changes declared physical content and therefore belongs to anchor or tensor structure. The two cases are separated by the same-scope boundary ledger. $\square$

22.1 Asymptotic and boundary time as a legitimate anchor

The boundary guard supplies a positive example of external time. In an asymptotically flat Hamiltonian formulation, differentiability of the generator requires surface terms, and an asymptotic time translation is paired with a physical charge such as the ADM energy [Regge and Teitelboim, 1974]. The associated parameter is not skin: the boundary conditions, generator, and charge belong to the theorem object.

Table 13: Bulk and boundary temporal data.

Temporal datum	Fixed-domain classification
Bulk coordinate time with no physical effect	Skin
Bulk lapse/shift variation at fixed boundary data	Presentation/redescription
Asymptotic time translation with nonzero Hamiltonian charge	Declared anchor and tensor content
Boundary clock used to condition observables	Boundary-trace or richer-scope clock package
External parameter used without boundary or state representation	Same-scope import failure

External time is legitimate when the physical boundary structure that carries it is declared and certified before the selection act. A boundary trace or temporal datum introduced in order to choose a preferred internal clock is scope enrichment, not same-scope fixation. Compact closed systems, asymptotically flat systems with boundary charges, AdS or other boundary-clock settings, and effective laboratory clocks therefore occupy different branches of the classifier. The theorem excludes only same-scope import of temporal authority absent from the theorem object.

Proposition 22.2 (AER classification of many-fingered evolution). *Within the bulk constrained comparison class, many-fingered hypersurface deformation is presentation/redescription structure.*

It becomes clock-bearing physical content only through relational observables, boundary generators, or records represented in the physical theorem object.

This proposition is not a universal claim that every interpretation treats refoliation as ordinary gauge. A theory that assigns distinct physical content to the deformations has changed the redundancy/continuation typing and must represent that content explicitly.

23 Matter clocks, dust time, and scalar time

Material reference systems can deparametrize gravity and provide physical readouts [Kuchař and Torre, 1991, Brown and Kuchař, 1995].

Proposition 23.1 (Relative-scope material-clock classification). *Relative to a gravity-only theorem object, adding reference matter is scope enrichment. Relative to a theory in which the matter degrees of freedom are already included, designating them as clocks may be an anchor or tensor assignment inside the existing scope. In either case, physical clock standing requires descent, a valid gauge-fixing/readout domain, and coherent continuation.*

The proposition avoids two errors: material clocks are not rejected because they add structure, and they are not reported as if their physical content emerged from vacuum gravity without that structure.

23.1 Scalar-field clocks

The worked audit in section 21 gives the local clock-standing certificate on its declared sector. The key point is relative scope: a scalar already present in the theory may receive an anchor/quotient-profile/tensor clock role on a monotonic patch; a new reference field is an explicit enrichment relative to a gravity-only object. In both cases, turning points, residual copies, physical-inner-product dependence, and readout failure remain part of the clock profile.

24 Page–Wootters and temporal quantum reference frames

The Page–Wootters construction represents relational evolution through correlations in a globally constrained state [Page and Wootters, 1983]. Schematically,

$$(H_C + H_S)|\Psi\rangle = 0, \quad |\psi_S(\tau)\rangle \propto \langle\tau|\Psi\rangle. \quad (50)$$

The displayed formula is kinematic shorthand. A physical implementation must specify the constraint solution space, physical inner product or rigging map, the clock observable/POVM, and normalization of the conditional state. Modern temporal-reference-frame work relates Page–Wootters conditioning, Dirac observables, and reduction maps, including nonideal and periodic clocks [Höhn et al., 2021].

Proposition 24.1 (Page–Wootters clock-standing certificate). *A Page–Wootters implementation supplies a physical clock-standing presentation only when:*

- (P1) *the clock-system split and interaction structure are declared;*
- (P2) *the global constraint and physical inner product are specified;*
- (P3) *the clock readout is a valid operator or covariant POVM on the relevant physical domain;*
- (P4) *conditional states are normalized and invariantly related to physical observables;*

(P5) *degeneracy, periodicity, finite dimension, nonideality, and backreaction are included in the failure profile;*

(P6) *any change of temporal reference frame preserves the full physical profile or is typed as anchor, quotient-profile/tensor, or theorem-object change.*

The parameter τ is then a representative label of relational tensor content, not primitive external time.

Table 14: Page–Wootters clock-standing certificate.

Certificate coordinate	Required physical datum
Constraint object	Physical stationary state and constraint implementation
Clock readout	Covariant clock POVM or appropriate operator family
Physical descent	Dirac-observable, reduction-map, or equivalent physical certificate
Conditioned state	Physical normalization and inner product
Clock quality	Finite dimension, periodicity, resolution, interaction, and backreaction
Clock change	Profile-preserving temporal-frame map or explicit anchor/tensor/scope change

The proposition fixes the certificate burden for Page–Wootters implementations. Application requires inspection of the physical Hilbert space, readout structure, conditional normalization, and failure domain. In implementations where equivalence to Dirac observables or reduction maps is established, that equivalence can provide the descent certificate. Where the clock is finite or periodic, a POVM and local/relational domain may replace an ideal self-adjoint time operator. The positive conclusion is substantive: Page–Wootters conditioning can realize physical clock standing without restoring primitive external time, provided the physical-state and readout certificate data are supplied. Selecting a clock subsystem is lawful only when the physical-state/readout package fixes the role; selection by convenience alone is representative privilege or unsupported class privilege. Two clock subsystems inducing the same conditional-state, readout, continuation, performance, and failure profile belong to one use-relative clock class. If they induce different conditional states or physical inner-product structures, the difference is quotient-profile divergence or theorem-object change. A Page–Wootters implementation realizes one or more canonical clock classes according to its certified conditional profiles; singular fixation requires a singleton quotient or declared governance route.

25 Thermal and modular time

For a von Neumann algebra $\mathcal{A}$ and a faithful normal state or suitable weight ω , Tomita–Takesaki theory supplies a modular automorphism group σ_t^ω . The thermal-time hypothesis proposes an operational temporal interpretation of that flow in generally covariant settings [Connes and Rovelli, 1994].

Proposition 25.1 (Thermal-time classification). *A modular flow is a physical clock candidate only relative to the declared algebra–state/weight package and an operational map from modular*

parameter to physical readouts or conditioned observables. It is tensor or anchor content of that package, not universal primitive time.

Table 15: Thermal-time role assignment.

Element	Clock-standing role
Von Neumann algebra and faithful state/weight	Declared anchor package
Modular automorphism group	Legitimate quotient-profile/tensor clock candidate
Operational readout or conditioned observable	Required physical-use certificate
Arbitrary choice of algebra or state	Unsupported class privilege if physically privileged without support
Globalization beyond the declared state/algebra domain	Scope transport or theorem-object change

The existence of a modular automorphism group is a mathematical result; its identification with physical clock readings is an additional physical interpretation. Selecting $\mathcal{A}$ or ω by convenience while treating the resulting flow as unique clock authority is unsupported class privilege or impossible authority acquisition unless the pair belongs to the declared theorem object. State dependence itself is not a defect when the state is part of the declared theorem object. Thermal time is therefore a legitimate quotient-profile/tensor candidate under a declared algebra–state and readout certificate; state relativity is part of the theorem object rather than a defect. A thermal-time construction realizes a clock class only relative to the declared algebra–state pair, operational readout role, and validity domain. Distinct algebra–state choices are canonically equivalent, anchor-divergent, theorem-object-changing, selection-governance dependent, or fail-closed; they do not constitute an untyped multiple-choice residue.

26 Histories, records, decoherence, and clock order

Consistent/decoherent-histories and records approaches seek temporal organization in correlations without presupposing a fundamental external parameter [Griffiths, 1984, Gell-Mann and Hartle, 1990, Halliwell, 1998]. Consistency or decoherence can support probabilities for history families, but does not by itself select a clock role or guarantee reusable records.

Proposition 26.1 (Histories clock-standing criterion). *A histories construction supplies a physical clock-standing presentation only when:*

- (H1) *a consistent/decoherent history family is declared;*
- (H2) *records for the relevant alternatives are fixed;*
- (H3) *one record role or relational ordering is assigned for the target use without representative privilege or unsupported class privilege;*
- (H4) *that order and the conditioned content descend to the physical quotient object;*
- (H5) *retained histories satisfy coherent continuation and a declared failure profile.*

Consistency or decoherence alone is insufficient.

A measurement record need not be a clock. A clock record is a record specifically used to order or condition other physical content. Everettian multiplicity does not alter the criterion: branch stability can provide candidate records, but a theory-level clock verdict still requires the base-record quotient and factorization test of [theorem 8.5](#). A histories construction realizes a clock class only when the history family, record algebra, ordering relation, reuse role, and failure domain are fixed. Families with the same record-order profile are one clock class; different decoherent families are anchor or quotient-profile/tensor divergences or theorem-object changes, while choosing one representative is representative privilege and choosing one non-equivalent family without declared support is unsupported class privilege.

27 Quantum cosmology and clock-dependent physical claims

A worked class of quantum-cosmology models shows that clock choice can affect unitarity, recollapse, singularity behavior, or the representation of relational evolution [[Gielen and Menéndez-Pidal, 2020](#)]. These results serve as concrete instances of the multiple-choice decision tree; the closure theorem does not depend on a universal generalization from any one model.

Proposition 27.1 (Clock-dependent verdict classification). *When two quantum-cosmology clocks yield different physical claims, their pair relation is theorem-object divergence, eligibility divergence, anchor divergence, quotient-profile/tensor divergence, or physical clock equivalence. If one candidate is then privileged, that privilege receives the separate governance classification of [theorem 15.3](#). No residual same-scope relation or selector category remains.*

For a minisuperspace scalar clock, nonzero momentum and monotonicity may support a local gauge fixing, but the physical inner product, positive-frequency sector, self-adjoint extension, and boundary conditions can also carry anchor or tensor load. A proper-time or scale-factor clock can therefore define a different theorem object rather than merely relabel one unchanged quantum theory. The AASC endpoint is closure of the admissible relation and selection routes; any representative preference requires declared authority.

Part V

The Physical Clock as a Quotient

Unique interior, quotient plurality

The fixed clock domain has one standing-governed admissible interior. The clock quotient formed inside that interior may nevertheless be empty, singleton, or plural. Clock plurality is typed physical content; plural admissibility regimes and undeclared selector authority are excluded.

28 Problem-of-time reclassification

Table 16: Standard facets and fixed-domain clock classification.

Facet	Diagnosis	Positive burden
Frozen formalism	No independently standing external generator on the qualified constraint target	Relational/conditional physical object, boundary Hamiltonian, or declared deparametrization
Multiple choice	Canonical quotient may be empty, singleton, or plural; pair relations and selection acts are distinct	Pair normal form, selection governance, and quotient-status classification
Observables	Bare clock labels do not define physical change	Descent to $\mathbf{Q}_D$ and native operator/domain control
Global time	One candidate fails outside a patch	Local validity plus compatible clock-atlas certificate
Foliation dependence	Representative slicing may be mistaken for physical content	Boundary guard and proof of skin neutrality, or declared anchor/tensor load
Hilbert/inner-product dependence	Clock choice may change the quantum theorem object	Explicit physical inner product, domains, and comparison map
Semiclassical time	Approximation package is hidden or overglobalized	Declared split, branch, gauge, order, and failure ledger
Records problem	Ordered traces are mistaken for a physical clock lineage	Base record quotient and conditioned-verdict factorization
Arrow confusion	Clock order, metric time, entropy increase, record asymmetry, and causal direction are collapsed into one object	Preserve the Paper I/Paper II distinction: standing arrow $<_S$, physical time T_D , effective arrows, and clock quotient $\mathcal{Q}_{\text{CLK}}(D, \mathbf{U})$ are distinct but compatible roles

The table is a constraint classification over native technical work. Every successful clock construction must discharge the positive burden in the formalism it uses, and no native mechanism is promoted into boundary authority merely by success.

29 Clock-neutral comparison across GR and quantum projections

The AER architecture treats canonical gravitational and quantum theorem objects as role-specific projections of a common admissibility-bearing quotient architecture [Maley, 2026a,j]. The present result adds one caution: temporal parameters in the two projections may be compared only through a common clock profile or an explicit transport certificate. Similar notation does not establish identity.

Proposition 29.1 (Clock-neutral comparison principle). *A GR-side temporal description and a quantum-side temporal description are projections of one clock architecture only when their use-relative profiles are images of one quotient-level class or are related by a map preserving anchors, physical observables, readouts, continuation, and failure typing.*

A clock choice cannot by itself supply primitive authority for a GR-to-quantum map. If it changes constraints, physical state space, inner product, boundary data, or observables, it is part of the realization architecture. If it changes none, it is skin and cannot carry physical force.

30 Referee route matrix

Table 17: Correctness burdens for the principal escape routes.

Objection	Required same-scope witness
Another clock functional exists	Preserve the full $(D, \mathcal{U})$ comparison class, carry standing-relevant force, and fail to factor through $\mathcal{H}_D^{\mathcal{U}}$.
The profile presentation is incomplete	Exhibit an admitted probe not captured by profile sufficiency and profile necessity.
Simplicity selects a preferred clock	Show that simplicity is declared anchor or quotient-profile content rather than representative privilege, unsupported class privilege, or carrier authority.
A plural clock quotient remains	Accepted as typed plurality inside the unique admissible interior; it does not constitute residual ambiguity or authorize selection.
A second clock-governance regime exists	Preserve the same gate and domain while escaping AMetric no-second-classifier closure and <i>Unus</i> no-discrete-interior closure.
Boundary time selects a clock	Show that the boundary datum was declared or independently admitted before selection; otherwise the move is scope enrichment.
Probability or typicality selects a clock	Escape AMetric no-parameter/no-selector closure and the no-carrier theorem without changing scope.
The scalar clock is globally preferred	Outside the theorem endpoint; the section establishes local realization, failure typing, and universal rival classification.

31 Certification output and reporting standard

The certificate and algorithm in [section 7](#) are the manuscript’s constructive output. A publication-grade clock claim should report the terminal classification together with the evidence supporting each certificate coordinate. For local clocks, overlap maps and patch failures should be explicit; for quantum clocks, the physical inner product, operator/POVM status, and normalization should be included; for effective clocks, approximation order and error control should be stated.

A certified clock claim therefore has the compact form

$$\left(\mathfrak{C}_D^{\mathcal{U}}, \mathcal{Q}_{\text{Cik}}(D, \mathcal{U}), [\Theta]_D^{\mathcal{U}}, \text{NF}_D^{\mathcal{U}}(\Theta), \text{Closed}_{\text{Cik}}(D, \mathcal{U}), \text{Sat}_D^{\mathcal{U}}(\mathcal{R}_D^{\mathcal{U}}) \right), \quad (51)$$

with a registry-completion status reported only when a finite application registry is actually used. The tuple records constraint status alongside model dynamics: the canonical clock quotient, candidate class, individual-candidate role, universal typing closure, and any additional registry-saturation burden.

32 Theorem boundary and falsifiability

The manuscript proves a full-strength fixed-domain clock-finality result. Its theorem object is clock-use closure rather than a universal quantum-gravity dynamics. Wheeler–DeWitt interpretation, global clock existence, construction of a physical inner product or Born rule, sector-specific material-clock and Page–Wootters dynamics, and the arrow of time belong to distinct or richer targets. Boundary Hamiltonians and non-gauge constraint interpretations remain typed by their declared theorem objects. In the series architecture, Paper II imports Paper I’s directionality distinction but neither replaces nor regenerates it.

It does claim, relative to the declared comparison class, that an external clock cannot be imported with same-scope physical force unless it is represented or declared in the theorem object; physical relational clock content must descend to $\mathcal{Q}_D$; theory-level conditioned verdicts must factor through the base clock-record quotient; the complete admitted probe family induces a final physical clock quotient, unique relative to the declared $(D, \mathcal{U})$ package; every clock pair receives one physical normal-form relation; every singular selection act receives one terminal governance status under the ordered theorem; failed clock patches are not repaired as the same lineage; and finite registry completion requires saturation.

Definition 32.1 (Faithful same-scope counterexample). A faithful counterexample preserves the same clock-use package, candidate class, quotient object, admitted probe family, use, record burden, continuation, and boundary data, and then supplies at least one of the following: (i) an admissible clock-use functional not constant on canonical clock classes; (ii) a quotient-respecting and separating presentation not isomorphic to $\mathcal{Q}_{\text{Clk}}(D, \mathcal{U})$; (iii) a physical clock-pair relation outside the five pair normal forms; or (iv) a same-scope clock-selection authority outside declared role fixation, prior boundary-trace fixation, scope enrichment, representative privilege, unsupported class privilege, and impossible authority acquisition. It uses no same-act repair or hidden change of theorem object.

Theorem 32.2 (Refutation criterion). *The full-strength closure package is refuted by a faithful survivor satisfying the preceding definition, by failure of clock-profile sufficiency or profile necessity, by failure of finality factorization, by an invalid imported theorem, or by a demonstrated same-scope selection route not covered by the governance theorem.*

A different clock model, richer theory, or denial of target instantiation does not refute the fixed-domain result unless it supplies one of those correctness witnesses. The burden remains: deny the target, break a load-bearing theorem, refute quotient finality, or exhibit the survivor.

33 Conclusion

The problem of time is not only a search for a temporal parameter. In every non-degenerate constrained domain that remains kernel-faithful, the clock problem is a quotient-finality and role-fixation problem for physical clock use. A reportable clock claim must identify its task, constrained domain, physical quotient object, readout and record ledger, continuation rule, performance-relevant content, and failure boundary. Those data determine a sufficient-and-necessary presentation of total admitted clock-use behavior.

The physical clock object is therefore not a privileged variable. It is the canonical quotient

$$\mathcal{Q}_{\text{Clk}}(D, \mathcal{U}) = \text{Cert}_D^{\mathcal{U}} / \sim_D^{\mathcal{U}}, \quad (52)$$

where equivalence means agreement under every licensed same-scope clock-use probe. The Clock-Profile Finality Theorem shows that every admissible present or future clock functional generated

by lawful continuation inside the fixed package factors through this quotient. The Clock-Quotient Uniqueness Theorem shows that every clock presentation satisfying profile sufficiency and profile necessity is canonically isomorphic to it.

Clock-pair relations and clock selection are distinct. Candidate pairs are theorem-object divergent, eligibility divergent, anchor divergent, quotient-profile divergent, or physically equivalent. Under the ordered governance theorem, an attempted singular selection is fixed by declared non-trace role content, fixed by a prior singleton boundary trace, scope-enriching, representative privilege, unsupported class privilege, or impossible authority acquisition. A Metric route collapse, *Unus* gate/interior uniqueness, and the Impossibility Suite close every residual selector, carrier, transfer, accumulation, repair, or bridge route. No same-scope clock identity, physical relation, or selector authority survives outside the quotient and declared anchors.

Relative to the declared package, the quotient may be empty, singleton, or physically plural inside one unique admissible interior. Empty means that the declared use has no certified clock. Singleton means one physical clock class, not one preferred representative. Plurality means genuine anchor or quotient-profile difference, not untyped ambiguity, plural gates, or plural admissibility regimes. The scalar-field audit illustrates local realization and failure without illicit global uniqueness; boundary time, WKB time, Page–Wootters conditioning, modular flow, material clocks, and histories occupy the same quotient-and-governance architecture when their native conditions are met.

The endpoint is consequently stronger than an audit list and narrower than a global solution of quantum gravity:

Relative to a declared (D, U) package, the physical clock is the canonical final quotient class of certified same-scope clock-use behavior. The multiple-choice problem reduces to the structure of that quotient—empty, singleton, or typed plurality inside one admissible interior—and no residual same-scope relation or selection authority survives outside the quotient and declared anchor structure.

The theorem closes clock choice rather than choosing by preference. A singular clock exists only when the quotient or declared governance fixes it; in plural regimes, nonselection is the final result rather than a gap. Clock identity, plurality, failure, and selection therefore stand only in the forms exhausted above.

Together with Paper I, the result is a two-sided non-primitivity theorem for time. The upstream kernel is instantiated by necessity in every non-degenerate standing-bearing regime. Paper I then forces standing monotonicity as the invariant directionality structure in kernel-faithful realized physical interiors. Paper II forces the final use-relative clock quotient in kernel-faithful constrained clock-use packages. Time is not removed: direction, metric representation, record order, causal and thermodynamic arrows, and clock use remain as distinct standing-governed structures inside admitted scopes. The AASC endpoint is clock choice closed by exhaustion, not resolved by an undeclared preference.

A Source-dependency and provenance index

The repository includes a machine-readable source manifest with exact filenames, version dates, imported theorem identifiers, source-version metadata, and SHA-256 hashes. The principal imports are summarized in [table 3](#). The manifest fixes the exact June 2026 source versions used by the dependency audit. Correctness depends on the imported statements and inferences, not on external source labels.

B Formal notation and object ledger

Table 18: Core notation.

Symbol	Meaning
D	Fixed constrained clock domain
U	Declared physical clock use
$\mathfrak{C}_D^U$	Declared clock-use package
Q_D	Physical quotient object in the native formalism
Θ_{form}	Formal relational clock candidate
Θ_{stand}^U	Physical clock-standing presentation
$\mathcal{H}_D^U$	Complete family of admitted same-scope clock-use probes
$\text{Pred}_D^U(h; \Theta)$	Definedness-and-value behavior of candidate Θ under probe h
$\Theta \sim_D^U \Phi$	Canonical equality of all admitted clock-use probe behavior
$\mathcal{Q}_{\text{Clk}}(D, U)$	Final canonical physical clock quotient $\text{Cert}_D^U / \sim_D^U$
$A_D^U(\Theta)$	Clock-use anchor/definedness signature
$P_D^U(\Theta)$	Invariant readout, record, continuation, performance, and failure profile
$[\Theta]_D^U$	Canonical physical clock class $q_D^U(\Theta)$
$\approx_{\text{ClkRec}}^0$	Equality of base clock-record profiles, excluding downstream verdict
V	Separately defined downstream conditioned verdict
$\text{ClockStand}_D^U(\Theta)$	Eligibility of Θ for U on D
NF_D^U	Ordered clock-candidate classifier
S	Attempted singular clock-selection act
$\text{Closed}_{\text{Clk}}(D, U)$	Universal pair-and-selection typing closure on the full candidate class
$\text{Sat}_D^U(\mathcal{R})$	Rival-registry saturation certificate
RegComplete_D^U	Saturated application-registry completion
Terminal classes	RichScope, Fail, Anchor, Tensor, and Skin

C Compressed proof of clock-quotient finality and choice closure

Fix D and U .

- (P1) Form the complete admitted probe family $\mathcal{H}_D^U$ from every licensed same-scope functional capable of changing clock-use classification, definedness, quotient output, readout/record behavior, coherent continuation, standing-relevant performance, or failure.
- (P2) Define $\Theta \sim_D^U \Phi$ by equality of all probe results in definedness and value. Constructional-behavior exhaustion ensures that any additional admissibility-relevant distinction is already a probe; a distinction changing neither classification nor retained continuation is skin.

- (P3) Prove profile sufficiency and profile necessity for the displayed anchor signature and invariant profile relative to $\sim_D^U$. ATS finality excludes unwitnessed profile refinements.
- (P4) Every admissible clock-use functional is constant on $\sim_D^U$ -classes and therefore factors uniquely through $\mathcal{Q}_{\text{Clk}}(D, U)$. This is clock-profile finality.
- (P5) Compare candidate pairs by theorem object, certification, anchor signature, and quotient class. Finality leaves exactly five physical pair relations.
- (P6) Audit any selection act separately. Declared non-trace role fixation and prior singleton boundary-trace fixation are positive routes; added selector structure changes scope; representative privilege, unsupported class privilege, and impossible authority acquisition fail.
- (P7) Therefore every pair relation and every selection act is typed. No residual same-scope clock identity, relation, or authority survives outside the canonical quotient and declared anchors.
- (P8) For finite application registries, add a saturation certificate before reporting completion.

D Comparison summary

Table 19: Relation to standard approaches.

Approach	Native contribution	Additional audit question
Kuchař–Isham analyses	Taxonomy of time and observable problems	Which proposed role is fixed, physical, and same-scope?
Partial/complete observables	Constraint-invariant relational predictions	Does the construction descend globally/locally and meet the declared use?
Reduced/deparametrized quantization	True Hamiltonian relative to a clock	Which clock, domain, inner product, and comparison map define the reduced theorem object?
Material reference systems	Physical clocks and coordinates from matter	Is reference matter already in scope or an explicit enrichment?
Page–Wootters/reference frames	Conditional dynamics from constrained states	Are physical inner product, POVM, normalization, and clock change certified?
Thermal time	Modular flow from algebra and state	What operational readout makes modular parameter a physical clock?
Histories/records	Temporal organization without external time	Are records fixed and reused in one quotient-stable clock role?
Semiclassical time	Approximate matter evolution on WKB background	Which split, branch, gauge, order, and failure domain are declared?

E Glossary

Table 20: Glossary.

Term	Meaning in this manuscript
Clock standing	Use-relative eligibility of a reference carrier for licensed physical reuse
Clock-use package	Domain, use, quotient, readout, record, continuation, and failure data fixing the target
Physical clock class	Canonical class in the final quotient $\mathcal{Q}_{\text{Clk}}(D, \mathcal{U})$, fixed by all admitted clock-use probes
Formal clock	Variable or construction used for relational/conditional observables without an assumed operational record burden
Physical clock presentation	Formal clock plus readout, record, continuation, and failure data for a specified use
Physical descent	Invariance under the declared redundancy relation and definition on $\mathcal{Q}_D$
Anchor	Same-domain precondition changing use definedness
Tensor	Quotient-level invariant load changing the physical clock profile on fixed anchors
Skin	Representative difference changing neither definedness nor physical profile
Richer scope	Terminal disposition for theorem-object change
Fail-closed	Terminal disposition for same-scope ineligibility
Base clock-record quotient	Quotient formed without the downstream verdict later tested for factorization
Clock atlas	Family of local physical clock presentations with certified overlap maps
Clock-profile finality	Every admissible same-scope clock-use functional factors through the canonical quotient
Clock-pair normal form	Five physical relations: theorem-object, eligibility, anchor, quotient-profile divergence, or equivalence
Clock-selection governance	Ordered classification of declared role fixation, prior boundary trace, scope enrichment, representative privilege, unsupported class privilege, and impossible authority acquisition
Clock-choice closure	Total typing of all candidate-pair relations and selection acts on the full same-scope class
Registry completion	Finite application closure only with a rival-class saturation certificate
Physical time role T_D	Ordered retained consequence inside the realized physical interior; not a preferred clock variable
Standing arrow	Irreversible strict order on standing classes; not identical to a clock, metric time, or an effective arrow
Effective arrow	Domain-specific record, thermodynamic, causal, quantum-measurement, psychological, or cosmological projection/refinement over standing monotonicity

Term	Meaning in this manuscript
No clock-to-kernel or clock-to-arrow backflow	The rule that certification or success of a clock carrier cannot generate the kernel, ground temporal direction, or authorize an effective arrow

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